

Beyond Prepare-and-Measure: The Cost of Trustless Detectors in Quantum Key Distribution

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Abstract

Quantum Key Distribution (QKD) was first put forward in 1984 by Bennett and Brassard through their foundational proposal for bipartite secure key exchange taking advantage of quantum mechanics, now commonly known as BB84. In the following decades, a number of inspired methods have emerged to guarantee shared secrecy. In particular, Measurement-Device-Independent (MDI) QKD, presented in 2012, improves BB84 security assumptions by eliminating the need to trust photon detectors. In order to be able to deploy QKD at scale, providers must map the trade-offs of either protocol, and comparisons of the two in secure information exchange rate (key rate), cost and scalability allow for efficient application of this maturing technology. In this report, we aim to contribute to the solution of this problem through use of NetSquid, a quantum network discrete event simulation (DES) engine, and an algebraic cost model, to simulate and compare network behaviour for both MDI-QKD and BB84. We see that the price to pay for MDI's improved security is a drop in key rate, of up to 10 times, in modest metropolitan networks under realistic hardware assumptions.

Contents

1	Introduction	4
2	Shared Quantum Secrecy	6
2.1	BB84	6
2.2	Measurement-Device-Independent QKD	7
2.3	Theoretical Performance Bounds	9
2.4	Network Topologies	9
3	Methodology	10
3.1	Performance Analysis	10
3.1.1	NetSquid Simulation Engine	11
3.1.2	Hardware Model	12
3.1.3	Bipartite Single-Link Analysis	14
3.1.4	Multi-User Network Analysis	16
3.2	Cost Model	19
3.3	Scalability Analysis	21
4	Results	22
4.1	Point-to-Point Performance	22
4.2	Network Performance	29
4.3	Cost	30
4.4	Deployment Recommendations	34
5	Discussion	37
5.1	The Security–Rate Trade-off	37
5.2	Cost Model Limitations	38
5.3	Simulation Model Limitations	39
5.4	Implications for Protocol Selection	42
6	Conclusion	44
6.1	Summary of Contributions	44
6.2	Findings	44
6.3	Future Work	45
A	Additional Point-to-Point Parameter Sweeps	48

B	Relay Placement Optimisation	50
C	Real-Topology Case Study: The BT London Network	56
D	NetSquid: Limitations and Speedup Strategies	60
E	Code and Reproducibility	63

1 Introduction

The advent of large-scale quantum computing poses a fundamental threat to classical public-key cryptography. Modern key-exchange standards, including RSA [1] and Elliptic Curve Cryptography [2, 3], derive their security from the computational hardness of integer factorisation and the discrete logarithm problem. Shor’s algorithm [4], published in 1997, demonstrated that both problems can be solved in polynomial time on a sufficiently powerful quantum computer, rendering these standards vulnerable to a quantum-capable adversary. Post-Quantum Cryptography (PQC) responds to this threat by identifying classically hard problems not susceptible to known quantum speed-ups; Quantum Key Distribution (QKD) takes the complementary route, using quantum mechanics to guarantee shared secrecy rather than computational assumption.

QKD was first conceived by Wiesner [5] in the 1970s and given its first operational form by Bennett and Brassard in 1984 [6]. In the decades since, QKD has matured into one of the most field-tested quantum technologies, with national-scale fibre deployments serving over 150 users across thousands of kilometres [7] and metropolitan testbeds demonstrating sustained multi-vendor operation [8, 9]. This trajectory has motivated a growing body of research into how QKD protocols should be chosen and deployed at scale. This work focuses on two protocols that represent a natural axis of comparison: BB84, the original Prepare-and-Measure scheme, and Measurement-Device-Independent QKD (MDI-QKD), proposed by Lo, Curty and Qi in 2012 [10]. Their protocols and theoretical performance bounds are introduced in Section 2.

MDI-QKD improves upon BB84’s security model by removing the need to trust photon detectors, an assumption that is difficult to verify in practice and has been exploited in hardware-level attacks. This advantage, however, comes at a real cost in key rate: at metropolitan distances, BB84 consistently delivers rates an order of magnitude higher than MDI under comparable hardware conditions [11, 12]. For network providers selecting between the two protocols, this trade-off must be mapped carefully across distance, topology, and network scale. The question is not simply which protocol is more secure, but under what deployment conditions MDI’s security advantage justifies its rate penalty, and at what economic and energetic cost.

Despite the commercial momentum behind both protocols — evidenced by a progression of field deployments from early national testbeds [7–9, 13, 14] through metropolitan and long-haul fibre networks [11, 15] to dedicated MDI-QKD systems [16–18], heterogeneous multi-vendor network trials [19], and European infrastructure programmes [20] — the quantitative basis for choosing between BB84 and MDI-QKD for a given application remains

poorly characterised. Simulation of MDI-QKD at metropolitan scale has not systematically explored the interplay between topology, relay placement, and hardware parameters across a sufficiently wide parameter space to support confident protocol selection [21]. Cost models for MDI networks are largely absent from the literature: whilst integer linear programming approaches have been developed for PM and entanglement-based QKD infrastructure [22], their extension to MDI has not been carried out. Energetic footprint, an increasingly central criterion in telecommunications infrastructure procurement, has only recently begun to receive attention in the QKD context [23]. A rigorous parametric comparison across key rate, cost, and scalability is absent.

This work addresses these gaps through a two-part methodology. First, we implement and simulate both protocols within NetSquid [24], a discrete-event quantum network simulation engine, across a ten-layer hardware model spanning ideal to physically realistic conditions. Second, we derive deployment cost formulae algebraically using symbolic per-component parameters—cost per photon source c_s , per single-photon detector c_d , and per km of installed dark fibre c_f —expressing each protocol’s total infrastructure cost as a closed-form function of user count and topology. Since manufacturer prices for quantum dot sources and single-photon detectors are not publicly available, absolute CapEx comparison is deferred to the parameter space; the structural scaling difference between protocols is the primary analytical result. NetSquid was chosen over a bespoke analytical or probabilistic model for a reason that outlasts this specific comparison: its component/node/protocol abstraction (Section 3) is general enough to host arbitrary quantum network topologies, hardware models, and protocols, not only BB84 and MDI-QKD. Classical networking has long relied on general-purpose discrete-event simulators such as ns-3 [25] and OMNeT++ [26] to plan and evaluate infrastructure at commercial scale; QKD deployment planning has no comparably mature counterpart. A secondary aim of this work is accordingly to lay early groundwork toward one, using this BB84-versus-MDI study as the first protocol-comparison workload the resulting infrastructure supports. We compare BB84 and MDI-QKD across three dimensions: key rate as a function of distance and hardware parameters; relative deployment cost as a function of network size; and scalability, defined as the marginal key rate when adding users to a fixed metropolitan topology. The remainder of this report is structured as follows. Section 2 introduces the protocols, their hardware underpinnings, and the theoretical performance bounds that govern them. Section 3 describes the simulation architecture and cost model. Section 4 presents results across all three comparison axes, and Section 6 concludes with deployment recommendations and directions for future work.

2 Shared Quantum Secrecy

Quantum Key Distribution refers to any communications protocol in which two or more parties use quantum channels, the sharing of quantum states, together with authenticated classical channels and post-processing, to establish a shared secret cryptographic key. The security of such a scheme is guaranteed by the laws of quantum mechanics, providing information-theoretic security against any eavesdropper whose capabilities are bounded by quantum physics. Since Bennett and Brassard’s foundational paper in 1984 [6], numerous variants of QKD have been proposed. This work focuses on two: the original Prepare-and-Measure scheme (BB84) and Measurement-Device-Independent QKD (MDI-QKD), proposed by Lo, Curty and Qi in 2012 [10].

2.1 BB84

BB84 is representative of the class of Prepare-and-Measure (PM) protocols. Alice begins by preparing a string of n photons, each encoded in one of four polarisation states (0° , 45° , 90° or 135°) selected uniformly at random. In the qubit formalism, these correspond to $|0\rangle$, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|1\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, respectively. Alice selects a basis ($\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$) for each qubit uniformly and independently, then uniformly and independently again, she applies a bit flip within that basis before transmission.

She transmits these quantum states to Bob over a quantum channel. Bob measures each photon in a uniformly and independently random basis. Alice and Bob then use an authenticated classical channel to compare basis choices and discard all photons measured in mismatching bases, producing a raw, sifted key. A portion of this sifted key is compared over the classical channel to estimate the Quantum Bit Error Rate (QBER): the fraction of photon pairs measured in the same basis that nevertheless disagree. A QBER below 11% is taken as evidence that the channel has not been compromised [27]. Alice and Bob then apply error correction and privacy amplification (hash-based post-processing that shortens the raw key, removing any partial information an eavesdropper may have gathered) to produce the final shared secret key. Figure 1 illustrates the basic bipartite logical layer.

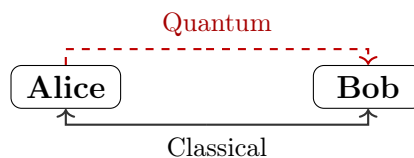


Figure 1: Communication diagram for point-to-point BB84. Alice sends quantum states to Bob; basis reconciliation and post-processing occur over authenticated classical channels.

Security proofs for BB84, and QKD generally, are conditional on the modelling assumptions made about devices, network, and attacker capabilities. A particular vulnerability in practice arises from weak coherent pulse (WCP) laser sources, whose photon number per pulse follows a Poisson distribution. Multi-photon pulses allow a Photon Number Splitting (PNS) attack [28], in which Eve intercepts and holds trailing photons until bases are announced, extracting key information without disturbing the QBER. The decoy-state technique [29] mitigates this by varying pulse intensity randomly, enabling Alice and Bob to tightly bound the single-photon contribution to their key.

2.2 Measurement-Device-Independent QKD

MDI-QKD addresses a fundamental vulnerability shared by all PM schemes: the need to trust the measurement devices at Bob’s node. Side-channel attacks on detectors represent a significant practical threat, as the gap between idealised device models assumed in security proofs and real hardware is difficult to close. MDI removes detectors entirely from the trust boundary by introducing an untrusted third-party relay, Charlie, who performs all measurement operations.

In MDI-QKD, both Alice and Bob independently prepare photon strings in the manner of BB84 and transmit these to Charlie. Charlie performs a Bell State Measurement (BSM) on each arriving photon pair, projecting onto the two-qubit Bell basis:

$$|\phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Charlie announces the measurement outcome over a classical channel to both Alice and Bob. Since the BSM measures only the *joint correlation* between the two incoming photons rather than their individual polarisations, an eavesdropper with full control of Charlie’s apparatus can learn nothing about Alice’s or Bob’s individual bit values. The physical mechanism underlying the BSM is Hong-Ou-Mandel (HOM) interference [30]: two indistinguishable photons arriving simultaneously at a 50:50 beam splitter always exit the same output port due to bosonic quantum interference. Coincident clicks at the downstream polarising beam splitters then identify the projected Bell state. Photon indistinguishability is therefore a prerequisite for MDI-QKD: timing jitter, spectral mismatch, and polarisation misalignment all reduce HOM visibility and degrade BSM success probability. The bipartite logical layer for MDI-QKD is shown in Figure 2.



Figure 2: Communication diagram for point-to-point MDI-QKD. Both Alice and Bob transmit quantum states to Charlie; Charlie performs a BSM and announces the result classically to both parties.

Alice and Bob sift by matching bases as in BB84. Upon receiving the BSM outcome, each party knows their own basis choice and bit value, and can deduce the other’s bit via the outcome. Depending on the Bell state announced, one party selectively bit-flips to ensure agreement, as summarised in Table 1. In the linear-optical BSM implementation, only two of the four Bell states ($|\psi^+\rangle$ and $|\psi^-\rangle$) are unambiguously distinguishable [10]; detection events corresponding to $|\phi^\pm\rangle$ are discarded as inconclusive. Privacy amplification and error correction then proceed as in BB84.

BSM outcome	$ \psi^+\rangle$	$ \psi^-\rangle$
Diagonal basis	No bit flip	Bit flip
Rectilinear basis	Bit flip	Bit flip

Table 1: Bit-flip rules upon receipt of BSM outcome [10]. Alice and Bob agree in advance which party applies the flip; if both flip, there is no net change.

The abstract operation of the BSM in quantum information terms is shown in Figure 3: a CNOT gate followed by a Hadamard and measurement. The physical implementation uses a 50:50 beam-splitter and polarising beam-splitters, with the Hong-Ou-Mandel effect ensuring that only coincidence detection events, two simultaneous clicks at the detector array, constitute valid BSM outcomes.

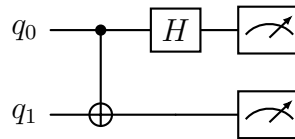


Figure 3: Bell State Measurement circuit: CNOT gate followed by Hadamard, then measurement in the computational basis.

MDI-QKD’s security advantage comes at a cost. Key generation requires a successful coincidence detection at Charlie – both photons from Alice and Bob must arrive and be detected within the same time window. This requirement, combined with independent connection losses on each arm, imposes a rate penalty relative to BB84’s single-link loss [10]. MDI’s QBER tolerance is also more sensitive than BB84’s, as two independent quantum channels each contribute noise, enlarging the attack surface accounted for by security proofs.

2.3 Theoretical Performance Bounds

The maximum achievable secure key rate in any point-to-point (no quantum repeater) QKD scheme is bounded by the Pirandola–Laurenza–Ottaviani–Banchi (PLOB) bound [31]. This fundamental result establishes that, against all attacks permitted by quantum physics, the key rate of a repeater-less link scales at most linearly with channel transmittance η :

$$R \leq -\log_2(1 - \eta) \approx \eta \log_2 e \quad \text{for } \eta \ll 1. \quad (1)$$

Since transmittance decays exponentially with fibre length L as $\eta = 10^{-\alpha L/10}$ (with α the fibre attenuation coefficient in dB/km), the PLOB bound implies that key rates must fall exponentially with distance. Both BB84 and standard MDI-QKD are subject to this bound.

Twin-Field QKD (TF-QKD), proposed in 2018 by Lucamarini et al. [32], exploits the same untrusted-relay tenet as MDI but replaces the two-photon BSM with single-photon interference. The result is that key rates scale as $\mathcal{O}(\sqrt{\eta})$ rather than $\mathcal{O}(\eta)$, exceeding the PLOB bound and extending the distance at which secure communication is achievable. TF-QKD thus represents the most practically significant near-term stepping stone beyond the repeater-less limit, and motivates the question central to this work, of precisely where and under what conditions MDI-QKD’s security advantage over BB84 justifies its rate penalty, before that regime is superseded by TF-QKD or, eventually, quantum repeaters.

2.4 Network Topologies

When QKD is deployed beyond a single point-to-point link, network topology becomes a primary driver of cost and scalability. BB84 requires a dedicated quantum channel between every pair of users: for N users, a full mesh demands $\binom{N}{2} = \mathcal{O}(N^2)$ fibre spans, each carrying an independent quantum signal. MDI-QKD’s architecture naturally admits a star topology instead: all users direct their pulses to a shared measurement node (Charlie), which performs the Bell-state measurement without accessing the key material. Crucially, Charlie need not be trusted—the measurement-device-independence property is preserved at network scale, so the relay introduces no new security assumption. For K relay nodes each serving a cluster of users, the MDI network requires $\mathcal{O}(N)$ user–relay fibre spans and $\mathcal{O}(K^2)$ inter-relay backbone links, both substantially below the $\mathcal{O}(N^2)$ full mesh for $K \ll N$. This structural asymmetry—summarised in the component-count model of Section 3.2—is the central economic and scalability argument evaluated in this work.

3 Methodology

We compare BB84 and MDI-QKD across three dimensions: performance, deployment cost, and scalability. Together these axes capture the trade-off a network provider faces when selecting between the two protocols for a given metropolitan deployment.

Performance. The primary metric is sifted key rate, measured in bits per second: the length of the matched-basis sifted key divided by the time taken to produce it. Since all QKD-generated keys are information-theoretically equivalent in security regardless of protocol, key rate is the sole performance discriminant between BB84 and MDI-QKD. We simulate key rate as a function of distance and individual hardware parameters using NetSquid [24], a discrete-event quantum network simulation engine, across a ten-layer hardware model that progressively introduces realistic physical imperfections from an ideal baseline.

Cost. Deployment cost is modelled algebraically using symbolic per-component parameters: c_s (photon source), c_d (single-photon detector), and c_f (fibre per km). The number of each component is determined by the network topology under each protocol. Since manufacturer prices for quantum dot sources and single-photon detectors are not publicly available at the precision required for quantitative comparison, we express total cost symbolically and focus on the *relative* cost structure between protocols. This reveals a structural crossover: BB84 fibre cost scales as $\mathcal{O}(N^2)$ whilst MDI scales as $\mathcal{O}(N)$ at fixed relay count K , so MDI becomes cost-competitive above a threshold user count N^* whose value depends on the ratio c_d/c_f and network geometry.

Scalability. We define scalability as the marginal key rate incurred when adding a user to an existing network. Since BB84 requires a direct link between every user pair whilst MDI routes traffic through relay nodes, the two protocols exhibit fundamentally different scaling behaviour. We quantify this by sweeping user count at fixed topology parameters and extracting per-user marginal key rates.

3.1 Performance Analysis

The primary performance metric is the sifted key rate R , measured in bits per second: the key length extracted by a simulation run divided by the simulation duration. Since all QKD-generated keys are information-theoretically equivalent in security regardless of protocol, key rate is the sole performance discriminant between BB84 and MDI-QKD for a given network. A symmetric QBER threshold of 11% is applied: simulation runs with QBER above this value are discarded as failures [27]. All key rates are computed in the asymptotic regime, without finite-key corrections, error correction, or privacy amplification — R is therefore

a sifted key rate, not a secure key rate (Section 5); this overestimates extractable secret key at low photon count rates but is standard practice in comparative simulation studies at metropolitan scale.

3.1.1 NetSquid Simulation Engine

NetSquid [24] is a discrete-event simulation (DES) engine for quantum networks developed at QuTech, chosen for this work for three reasons. First, NetSquid natively tracks qubit states as density matrices and exposes standard gate operations (`ns.X`, `ns.H`, `ns.CNOT`) and projective measurement (`ns.qubits.measure`), enabling BB84 encoding and the MDI Bell state measurement to be implemented directly as quantum operations rather than hand-coded probability tables. Second, the DES clock provides per-trial simulation timestamps, making key rate computable from the elapsed simulation time between photon emission and protocol completion; propagation delay is introduced naturally by the channel length parameters of `QuantumChannel` and `ClassicalChannel` objects via a `HybridDelayModel`. Third, the `NodeProtocol` abstraction models each party’s logic as an event-driven coroutine that yields at `await_port_input` calls until a message arrives; this allows Alice, Bob, and Charlie’s asynchronous exchanges (photon emission, index tagging, BSM announcement, basis sifting) to be written sequentially without explicit thread management, and maps directly onto the protocol descriptions in Section 2.

NetSquid organises a simulation into three layers of abstraction. *Components* represent physical hardware: `QuantumChannel` objects carry photons between nodes, applying noise models (loss, dephasing, delay) as Kraus operators; `ClassicalChannel` objects carry classical messages for basis reconciliation and BSM announcements. *Nodes* represent network parties (Alice, Bob, Charlie), each holding named ports through which components connect; nodes own their local quantum processors and memory registers. *Protocols* are event-driven coroutines that run on nodes, yielding control to the DES scheduler at each `await` expression until a subscribed event fires. This three-layer structure is modular: changing a component’s noise model is localised to a single object and does not require protocol logic to change. The same separation is what makes NetSquid a plausible foundation for a general-purpose QKD network simulator rather than a single-comparison tool: adding a further protocol, hardware model, or topology means adding new Components, Nodes, and Protocols alongside the existing BB84 and MDI-QKD implementations, not restructuring the engine underneath them. This is the same design principle underlying commercial-grade classical network simulators such as ns-3 [25] and OMNeT++ [26], and it is the property this work leans on in treating NetSquid as a starting point for an analogous QKD-specific tool (Section 6). NetSquid’s computational cost, and the extent to which this general-purpose engine is well suited to

Layer	Parameter added	Symbol	Default value	Verification source
0	Ideal baseline	—	All perfect	—
1	Fibre attenuation	α	0.18 dB/km	Corning [33]
2	Detector efficiency	η_Z	0.90	ID Quantique [34]
3	Dark count rate	d_c	50 cps	ID Quantique [34]
4	TX insertion loss	L_i	0.10 (10%)	OZ Optics [35]
5	RX node loss	L_n	2.0 dB	Duplinskiy <i>et al.</i> [36]
6	Source error rate	ε_s	0.015	Quandela [37]
7	Fibre dephasing	β	$3.2 \times 10^{-7} \text{ km}^{-1}$	Corning [33]
8	X-basis detector bias	η_X	0.715	Grasselli <i>et al.</i> [38]
9	Beam-splitter efficiency	η_{BS}	0.97	Thorlabs [39]
—	Relay position	δ	0.5	User symmetry

Table 2: Ten-layer hardware model. Layer k includes all impairments of layer $k - 1$ plus the new parameter listed. Layer 9 is the realistic operating point used for network simulations. Empty cells in the verification column indicate parameters pending datasheet confirmation.

network-scale QKD simulation, is discussed in Appendix D.

3.1.2 Hardware Model

Physical impairments are introduced via a ten-layer parametric model. Each layer adds one impairment cumulatively to the previous, from a fully ideal baseline (layer 0) to a system incorporating all realistic imperfections (layer 9). This structure allows the marginal impact of each physical parameter on key rate to be isolated, forming the basis of the parameter sweep analysis in Section 4. Table 2 lists the ten layers.

Layer 0 is the ideal baseline: photons propagate without loss or noise, so key rate is limited only by propagation delay. Bipartite sweeps at this layer vary distance across the metropolitan range (0–50 km), establishing an upper bound on achievable key rate and confirming correct simulation timing before physical impairments are introduced.

Layer 1 adds fibre attenuation: each photon is discarded before transmission with probability $p_{\text{loss}} = 1 - 10^{-\alpha L/10}$, where $\alpha = 0.18 \text{ dB/km}$ and L is the arm length. For MDI-QKD the two arms attenuate independently, so coincidence probability scales as the product of both arm transmittances.

Layer 2 introduces Z-basis detector efficiency $\eta_Z = 0.90$: each photon arriving at the receiver is accepted with probability η_Z . Missed detections reduce key length but do not introduce errors.

Layer 3 adds dark count rate $d_c = 50 \text{ cps}$: a spurious click fires per time slot with probability $d_c/(d_c + f)$, where f is the source repetition rate. Dark clicks on empty slots generate false detections; dark clicks on occupied slots override the measurement with a

random bit, contributing directly to QBER.

The default detector parameters ($\eta_Z = 0.90$, $d_c = 50$ cps, layers 2–3) correspond to a superconducting nanowire single-photon detector (SNSPD) operating point, verified against the ID Quantique (IDQ) ID281 [34]. The bipartite parameter sweeps in Section 4.1 additionally span the single-photon avalanche diode (SPAD) regime ($\eta_Z \approx 0.10$ – 0.25 , $d_c \approx 100$ – 200 cps), representative of the IDQ ID230 [40], covering the full range of detector technologies relevant to near-term metropolitan deployments.

Layer 4 adds TX insertion loss $L_i = 0.10$: a fixed fraction of photons is lost at the source before entering the fibre, absorbed into the combined transmittance $T = 10^{-\alpha L/10}(1 - L_i)$. This models a polarisation-maintaining fused fibre coupler combining two orthogonally-polarised branches (e.g. H/V or $\pm 45^\circ$) into a single output fibre for state preparation. Since the combined signals differ in polarisation, the coupler avoids the reciprocal 3 dB splitting loss that would apply to identical inputs [35]; only the excess loss is incurred. Taking the OZ Optics DTS0092 PM fused coupler’s maximum excess loss of 0.4 dB, $L_i = 1 - 10^{-0.4/10} \approx 0.09$, rounded up to 0.10 for a safety margin.

Layer 5 introduces receiver node loss $L_n = 2.0$ dB as a single lumped parameter applied uniformly to both protocols, capturing the insertion loss of optical receiver components (EOM, PBS, HOM coupler). Each photon surviving fibre transmission is discarded with probability $1 - 10^{-L_n/10}$ before reaching the detector. This value matches the measured total insertion loss of Bob’s receiver stage (LiNbO₃ phase modulator for basis selection, followed by a polarising beamsplitter) in a demonstrated all-fibre polarisation-encoded BB84 system [36], giving $\eta_{\text{node}} = 10^{-2.0/10} \approx 0.63$. Individual component losses are not modelled separately and the protocol-specific breakdown is noted as a limitation in Section 5.

Layer 6 adds source error rate $\varepsilon_s = 0.015$: each emitted photon receives an unintended $\hat{X}$ flip with probability ε_s , modelling multi-photon emission events characterised by $g^{(2)}(0)/2$ for the reference single-photon source [37]. The Quandela Prometheus datasheet specifies a second-order correlation $g^{(2)}(0) < 0.03$: the probability that a given pulse contains a multi-photon emission rather than a single photon. Taking the standard approximation that half of such multi-photon events go undetected as errors, $\varepsilon_s \approx g^{(2)}(0)/2 < 0.015$. Ideal single-photon sources are assumed throughout; WCP sources and decoy-state techniques [29] are not modelled, so reported rates represent an upper bound relative to practical laser deployments.

Layer 7 adds fibre dephasing $\beta = 3.2 \times 10^{-7}$ km^{−1}: the `DephaseNoiseModel` applies a per-photon dephasing error with probability $p_\phi = \min(1, \beta L)$ proportional to arm length, modelling polarisation-mode dispersion in standard single-mode fibre. This value is derived from the Corning SMF-28 PMD coefficient $D_{\text{PMD}} = 0.04$ ps/ $\sqrt{\text{km}}$ [33]: mean differential group delay $\Delta\tau = D_{\text{PMD}}\sqrt{L}$ induces a polarisation-dephasing probability $p_\phi \approx (\Delta\tau/T)^2/2$,

with $T = 50$ ps the detector timing-discrimination window (IDQ ID281 jitter [34]). Since $\Delta\tau^2 \propto L$, p_ϕ is linear in L , giving a constant per-km rate $\beta = D_{\text{PMD}}^2/(2T^2)$. Deployed fibre is assumed to vary $\pm 20\%$ around this value (2.6×10^{-7} – 3.8×10^{-7} km $^{-1}$) to reflect manufacturing spread within the SMF-28 specification.

Layer 8 introduces X-basis detector bias $\eta_X = 0.715$: photons measured in the X basis are detected with lower efficiency than those in the Z basis, reflecting the extra optical stage (dispersive medium/interferometer) required for the test-basis measurement. This value follows Grasselli *et al.* [38], who model the test-basis (X) measurement’s extra optical stage as an additional 1 dB insertion loss, giving $\eta_X = \eta_Z \times 10^{-1/10}$. Since insertion loss is a passive multiplicative factor applied upstream of the detector, this ratio is independent of the detector’s own quantum efficiency and applies directly at $\eta_Z = 0.90$: $\eta_X = 0.90 \times 10^{-1/10} \approx 0.715$.

Layer 9 (MDI only) adds beam-splitter efficiency $\eta_{\text{BS}} = 0.97$: the HOM coupler at Charlie’s relay has non-unit splitting ratio, reducing coincidence detection probability per arm. This value is derived from the Thorlabs TN1550R5F2 fibre-coupled 50:50 coupler [39], a 1550 nm narrowband fused coupler specified with excess loss < 0.15 dB: $\eta_{\text{BS}} = 10^{-0.15/10} \approx 0.97$. The relay position $\delta = 0.5$ (Charlie at the link midpoint) is held fixed for all bipartite sweeps, reflecting user symmetry; it is determined geometrically in the network setting via Equation 9.

3.1.3 Bipartite Single-Link Analysis

The bipartite simulation sweeps one hardware parameter at a time, holding all others at their layer-9 defaults (Table 2). For each parameter value, $r = 20$ independent Monte Carlo trials are run; each trial constitutes one complete QKD exchange from photon emission through key sifting. The mean key rate, mean QBER, and success rate (fraction of trials with QBER $< 11\%$) are reported as a function of the swept parameter.

BB84 simulation. Alice and Bob are represented as `Node` objects connected by one quantum channel (Alice $\rightarrow$ Bob) and two classical channels for basis announcement and sifting. Alice prepares $n = 1,024$ photons at $f = 10^7$ Hz. Each photon is assigned an independent random basis (Z or X) and bit value. Encoding is applied qubit-by-qubit: a bit-1 value applies $\hat{X}$; an X -basis choice applies a Hadamard gate. Source imperfection (ε_s) is modelled by an additional $\hat{X}$ flip applied with probability ε_s , consistent with the multi-photon emission rate $g^{(2)}(0)/2$ of the reference single-photon source. Photons are discarded before transmission with probability

$$p_{\text{loss}} = 1 - 10^{-\alpha L/10} (1 - L_i), \quad (2)$$

modelling both fibre attenuation and TX coupling loss. The quantum channel applies an additional per-photon dephasing error of probability βL . At Bob, each arriving photon is further subject to receiver node loss (probability $1 - 10^{-L_n/10}$), then detected with probability η_Z (Z basis) or η_X (X basis). Dark counts fire with per-slot probability $d_c/(d_c + f)$; a dark click on a slot where a real detection also occurs overrides with a random outcome. Dark clicks also fire on slots where the photon was lost.

After transmission, Bob announces his measurement bases and the indices of his detected slots; Alice returns her basis list. Bob retains only detections where both parties chose the same basis. The sifted key is the list of Bob's raw measurement outcomes at matching-basis slots. QBER is the fraction of bit disagreements between Alice's and Bob's sifted keys:

$$\text{QBER} = \frac{1}{\ell} \sum_{k=1}^{\ell} \mathbf{1}[a_k \neq b_k], \quad (3)$$

where ℓ is the sifted key length. The QBER is measured over the full sifted key, not a sacrificed subset, since no eavesdropper is modelled. Key rate is then:

$$R = \frac{\ell}{T_{\text{end}} - T_{\text{start}}} \quad (\text{bps}), \quad (4)$$

where T_{end} is the simulation time at which Bob completes basis reconciliation and T_{start} is the photon emission start, both in nanoseconds. Trials with $\text{QBER} > 11\%$ are excluded and counted as failures. No error correction or privacy amplification is applied; R is therefore a sifted key rate rather than a secure key rate, and overestimates the latter (see Section 5).

MDI simulation. Three nodes are instantiated: Alice, Bob, and Charlie. Two quantum channels (Alice $\rightarrow$ Charlie, Bob $\rightarrow$ Charlie) and six classical channels (BSM announcement and basis disclosure) are constructed. Each end node prepares $n = 512$ photons — half the BB84 count — normalising comparison to a single-source rate scale, since MDI involves two independent emitters per trial. Encoding, source error, and per-photon transmission loss are applied identically to the BB84 case, with arm lengths $L_A = L\delta$ and $L_B = L(1 - \delta)$ where $\delta = \text{charlie_pos}$ locates Charlie on the link; each arm applies an independent dephasing model proportional to its length.

At Charlie, photons arriving from Alice and Bob on the same pulse index are paired. Both photons undergo node coupling loss (L_n) and are detected with probability $\eta_{\text{BS}} \eta_d$ (HOM beam-splitter efficiency $\times$ detector efficiency). A CNOT gate followed by a Hadamard on the control qubit and measurement of both qubits implements the Bell state measurement.

The four computational outcomes correspond to the four Bell states:

$$(a, b) \mapsto \begin{cases} |\Phi^+\rangle & a = 0, b = 0 \\ |\Phi^-\rangle & a = 1, b = 0 \\ |\Psi^+\rangle & a = 0, b = 1 \\ |\Psi^-\rangle & a = 1, b = 1 \end{cases} \quad (5)$$

Consistent with the linear-optical BSM constraint [10], only $|\Psi^\pm\rangle$ outcomes ($b = 1$) are treated as conclusive; $|\Phi^\pm\rangle$ outcomes ($b = 0$) are discarded. This yields a maximum BSM success probability of 50% for coincident pairs. Dark count clicks on unpaired or lost slots generate spurious conclusive outcomes, contributing to QBER. Charlie announces BSM outcomes and basis-mismatch indices to Alice and Bob; end users discard inconclusive and basis-mismatched rounds. Bob applies basis-dependent bit corrections according to the BSM outcome to restore key agreement with Alice. The sifted key, QBER, and key rate are computed as per Equations 3–4, with $T_{\text{end}} = \max(T_{\text{end}}^A, T_{\text{end}}^B)$. In the network setting, the Charlie position is determined geometrically by the relay’s location relative to the user pair (Section 3.1.4).

3.1.4 Multi-User Network Analysis

The multi-user simulation models a metropolitan network of N users within an $A \times A$ km service area. Users are placed uniformly at random within the grid.

Relay placement. Placing the K MDI relay nodes is a facility-location problem: given fixed user locations and free relay nodes that connect to their assigned users and, for $K > 1$, to each other over a fully meshed backbone (Table 3), the placement that minimises deployment cost is the one that minimises total fibre length, since fibre enters the cost model linearly through c_f (Section 3.2; Equation 11). This is a sum of *unsquared* distances, so the objective that actually matches the network’s linear cost structure is the geometric median (the Weber point), not the centroid a plain k -means returns, which instead minimises squared distance; the two coincide only for symmetric configurations. The single-facility geometric median has no closed form and is obtained iteratively via Weiszfeld’s algorithm,

$$x^{(t+1)} = \frac{\sum_i p_i / \|x^{(t)} - p_i\|}{\sum_i 1 / \|x^{(t)} - p_i\|}, \quad (6)$$

iterated to convergence.

For $K > 1$ fully-meshed relays, the placement problem does not decompose into K independent per-cluster geometric medians, because the inter-relay backbone term couples all relay positions together. Writing cluster k 's user set as C_k and its relay position as x_k , total fibre length is

$$L(x_1, \dots, x_K) = \underbrace{\sum_{k=1}^K \sum_{i \in C_k} \|x_k - p_i\|}_{\text{spoke cost}} + \underbrace{\sum_{k < l} \|x_k - x_l\|}_{\text{backbone (mesh) cost}}. \quad (7)$$

L is jointly convex in $(x_1, \dots, x_K)$, so a unique global minimum exists, but setting each x_k independently to its own cluster's geometric median ignores the mesh term and is generally suboptimal. Zeroing $\nabla_{x_k} L$ instead gives the generalised (multi-facility) Weiszfeld fixed-point update, with convergence to the joint optimum guaranteed by convexity [41]:

$$x_k^{(t+1)} = \frac{\sum_{i \in C_k} \frac{p_i}{\|x_k^{(t)} - p_i\|} + \sum_{l \neq k} \frac{x_l^{(t)}}{\|x_k^{(t)} - x_l^{(t)}\|}}{\sum_{i \in C_k} \frac{1}{\|x_k^{(t)} - p_i\|} + \sum_{l \neq k} \frac{1}{\|x_k^{(t)} - x_l^{(t)}\|}}, \quad (8)$$

updating all K relays simultaneously each iteration. Appendix B derives this update from first principles, works a small example illustrating why it diverges from the k -means centroid, and reports a geometric comparison of the two placement methods across synthetic topologies, quantifying the resulting fibre-length reduction.

This work places relays by the generalised Weiszfeld iteration of Equation 8, in place of a plain k -means centroid. Cluster assignment (spoke membership C_k) is still initialised by k -means with $n_{\text{init}} = 10$ random restarts, a reasonable heuristic for partitioning users spatially; relay *position* within each cluster is then refined by Equation 8 rather than fixed at the cluster centroid, and assignment is re-evaluated against nearest-relay distance after each position update, alternating position refinement and reassignment to convergence. For $K = 1$ this reduces to the plain single-facility Weiszfeld iteration of Equation 6, since the mesh term is absent; the joint coupling in Equation 8 has an effect only for $K > 1$. User-to-relay assignment remains $\text{user_relay}[i] = \arg \min_k \|u_i - r_k\|_2$, evaluated against the converged relay positions. This is a methodological change from earlier k -means-centroid placement used elsewhere in this work's development. It is used exactly as described here — k -means-seeded, Weiszfeld-refined, with membership re-evaluated by nearest-relay distance after every position update — specifically for the relay count sweep of Section 3.3, where relay infrastructure is genuinely redesigned from scratch at each K (a greenfield deployment decision). The user count sweep of Section 3.3 and the relay-placement strategy comparison of Appendix B instead use the same

position update (Equation 8) with catchment membership held *fixed* rather than reassigned: those experiments specifically test how well a placement decision made once serves users it was never re-optimised for, which the reassigning variant would undermine by silently re-clustering around whatever the relay’s current position happens to be. Appendix B justifies that choice in detail. The geometric-median objective could still revert to a plain centroid in either variant if the fibre-length cost model is later judged not to warrant the added iteration.

To decouple topology variability from Monte Carlo noise, all network experiments are repeated across S independently drawn user placements (seeds), with relay positions re-optimised per seed. Aggregate statistics across seeds produce error bars that characterise topology uncertainty — the variance arising from user placement — rather than the sampling noise of a fixed topology. The number of seeds is chosen such that the inter-seed standard deviation of the protocol comparison metric (mean key rate) is small relative to the protocol gap.

This work focuses on metropolitan-scale networks, where inter-node distances are short enough for both BB84 and MDI-QKD to establish keys at viable rates. Two network protocols are simulated:

BB84 mesh. Every user pair (i, j) communicates over a dedicated direct dark-fibre link. Installed cable follows duct routes longer than the straight-line Euclidean distance; per-link tortuosity is sampled independently for each of the $N(N-1)/2$ links from a truncated normal distribution, $\tau_{ij} \sim \max(1, \mathcal{N}(\mu_\tau, \sigma_\tau^2))$ with $\mu_\tau = 1.2$, $\sigma_\tau = 0.1$, representative of urban cable routing where installed fibre is typically 15–25% longer than the Euclidean distance. The physical cable length is $d_{ij} \tau_{ij}$ km. Tortuosity affects both the simulated key rate, through increased fibre loss on longer cables, and installed fibre cost (Section 3.2). All pairs are dispatched to the multiprocessing pool simultaneously; r Monte Carlo trials are run per pair. Network key rate is the mean over all valid (pair, trial) combinations.

MDI network. Tortuosity is sampled once per physical cable rather than per pair, ensuring that all simulation pairs sharing a cable see the same physical routing factor. Specifically, N user-to-relay cables and $K(K-1)/2$ relay-to-relay backbone cables each draw one independent tortuosity factor at the start of each seed.

For a same-cluster pair (i, j) assigned to relay $R_i = R_j = R$, Alice’s arm length is $d_{iR} \tau_{iR}$ and Bob’s arm length is $d_{jR} \tau_{jR}$; the BSM takes place at relay R . For a cross-cluster pair (i, j) with $R_i \neq R_j$, Bob’s photon travels from user j to relay R_j , then along the backbone to relay R_i where Alice’s BSM station is located. Bob’s effective arm length is $d_{jR_j} \tau_{jR_j} + d_{R_j R_i} \tau_{R_j R_i}$. The Charlie position parameter is set geometrically:

$$\delta = \frac{d_{iR_i} \tau_{iR_i}}{d_{iR_i} \tau_{iR_i} + d_{jR_j} \tau_{jR_j} + d_{R_j R_i} \tau_{R_j R_i}}. \quad (9)$$

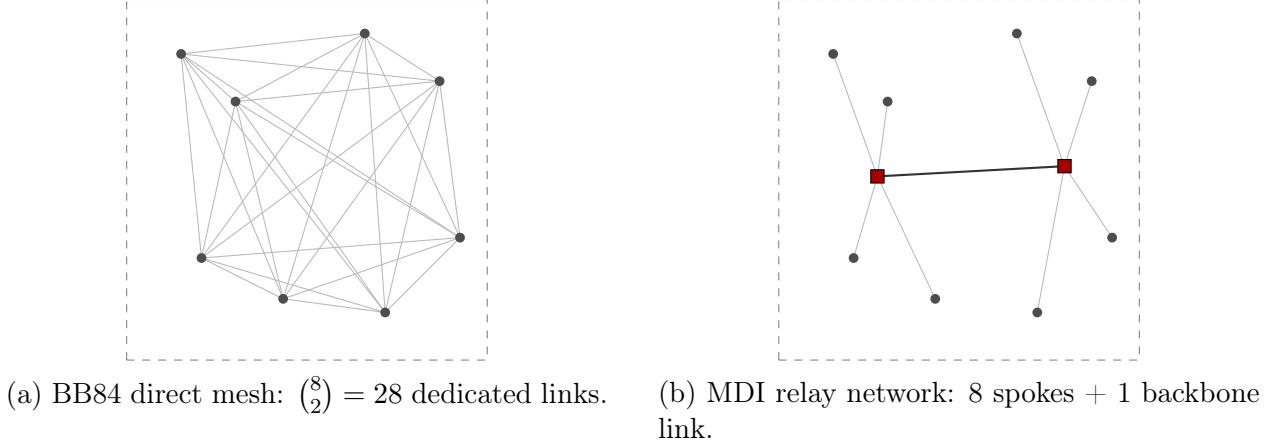


Figure 4: Example network topology for the same $N = 8$ user configuration under BB84 (left) versus MDI-QKD with $K = 2$ relays (right, relay stations shown as squares). BB84 requires a dedicated fibre for every user pair ($\mathcal{O}(N^2)$ links); MDI instead routes each user to its nearest relay, with the two relays joined by a shared backbone link ($\mathcal{O}(N)$ spokes plus $\mathcal{O}(K^2)$ backbone links). Relay positions here are illustrative cluster centroids, not the geometric-median placement of Section 3.1.4.

An optical switch at each relay routes backbone photons to the BSM station for cross-cluster pairs. Its insertion loss is set to 0 dB in the current model; the asymmetry between switch installation cost (per relay) and switch-induced loss (cross-cluster pairs only) is noted as a simulation limitation in Section 5.

The network key rate reported throughout this work is the mean pairwise point-to-point key rate averaged over all $N(N-1)/2$ user pairs and S topology seeds. This is not a live traffic simulation: no channel contention, scheduling, or multiplexing is modelled. Every user pair is assumed to extract key simultaneously and independently at full channel capacity, with no interference from other pairs sharing physical infrastructure. The reported metric therefore constitutes a semi-probabilistic geometric upper bound on network throughput: *geometric* because the rate for each pair is determined by the physical path geometry (distances, relay placement, tortuosity); *probabilistic* because relay positions and cable tortuosities are drawn stochastically and averaged across seeds; and an *upper bound* because real deployments would see contention for shared fibre, detector dead time across concurrent channels, and scheduling overhead that this model does not capture. The multi-seeding procedure described above quantifies the geometric component of this bound across the distribution of plausible topologies for a given N , K , and service area.

3.2 Cost Model

We model the total deployment cost of a QKD network algebraically using three symbolic per-component parameters: c_s (cost per photon source), c_d (cost per single-photon detector),

and c_f (cost per km of installed dark fibre), since manufacturer prices for quantum dot photon sources and single-photon detectors are not publicly available. All costs are capital expenditure (CapEx) only; operational expenditure — in particular the cryogenic cooling required by superconducting nanowire single-photon detectors (SNSPDs) and quantum dot (QD) sources — is not modelled. Classical communication infrastructure (basis reconciliation, error correction, and key forwarding) is assumed to be a sunk cost of existing network infrastructure. The most directly related prior cost analysis is that of Karavias *et al.* [22], which applies integer linear programming to PM-QKD and entanglement-based infrastructure planning with concrete component costs; MDI-QKD is not considered there. We extend this structural analysis to MDI-QKD; the resulting cost formulae, crossover analysis, and commercial packaging discussion are presented alongside the results they inform in Section 4.3.

The two network topologies impose fundamentally different hardware structures. In BB84, every user pair requires a dedicated direct dark-fibre link: the network is a complete mesh of $N(N-1)/2$ quantum channels. At each receiver, a polarising beam splitter (PBS) separates Z and X basis outputs to two single-photon detectors (SPDs), and an electro-optic modulator (EOM) performs active basis selection. No relay infrastructure is required.

In MDI-QKD, users transmit photons to BSM relay stations rather than to each other. Each relay is a passive station comprising a 50:50 HOM beam splitter, two PBS (one per arm), four SPDs, and one optical switch for routing photons from cross-cluster pairs. The network requires N user-to-relay fibre links and $K(K-1)/2$ relay-to-relay backbone links.

Table 3 summarises the component counts for both protocols.

Component	BB84	MDI
Single-photon sources	N	N
Single-photon detectors	$2N$	$4K$
50:50 beam splitter (HOM)	0	K
Polarising beam splitter	N	$2K$
Electro-optic modulator	N	0
Optical switch	0	K
Fibre links	$N(N-1)/2$	$N + K(K-1)/2$

Table 3: Hardware component counts for a network of N users and K relay nodes. MDI relay nodes are passive BSM stations. BB84 forms a direct mesh with no relay infrastructure.

Every fibre link’s installed cost is linear in its length via c_f (Section 3.1.4 uses this to justify a fibre-length placement objective), and fibre length itself scales as $\mathcal{O}(N^2)$ for BB84’s complete mesh versus $\mathcal{O}(N)$ for MDI at fixed relay count K (Table 3) — the structural cost

incentive for relay-based architectures as user counts grow, and the difference the full cost formulae of Section 4.3 build on to derive a cost crossover user count N^* . Installed fibre length additionally reflects the per-link tortuosity model of Section 3.1.4.

3.3 Scalability Analysis

Scalability is characterised through two orthogonal sweeps.

User count sweep. Rather than generating the full $N_{\max}$ -user network up front and revealing a prefix of it, users are added one at a time from a single random stream: each new user joins a uniformly-random existing relay catchment and is placed by an isotropic Gaussian around that catchment’s centre, so that $N \rightarrow N+1$ genuinely means *adding* a user rather than redrawing an unrelated sample. Relay positions are placed once, for the $N_{\min}$ -user network, by whichever placement strategy is specified (centroid, boundary, or the Weiszfeld iteration of Equation 8), and then held fixed as N grows to $N_{\max}$; catchment membership, once assigned, is likewise never reassigned by nearest-relay distance. This models a provider that sizes relay infrastructure for an initial deployment and serves subsequent organic growth without continuously re-optimising it, rather than a provider that instantly re-plans infrastructure after every new customer. For each N , both protocols are simulated and their mean key rate is recorded. The marginal rate $\Delta\bar{R}/\Delta N$ is computed numerically from adjacent sweep points. The algebraic cost model separately predicts that MDI marginal CapEx decreases with N (relay hardware is sunk, only user-to-relay fibre grows linearly), whilst BB84 marginal CapEx grows as $\mathcal{O}(N)$ since each new user requires $N - 1$ new direct fibre links.

Relay count sweep. User count N is fixed and K is varied from 1 to $K_{\max}$. BB84 is relay-independent and appears as a constant baseline. MDI key rates improve as K increases (shorter mean user-to-relay distances) until the relay-to-relay backbone becomes the dominant bottleneck. This sweep surfaces the performance-optimal relay count for a given N and area.

4 Results

4.1 Point-to-Point Performance

Eleven independent parameter sweeps characterise bipartite BB84 and MDI-QKD key rate around a common metropolitan operating point set by the values in 2, with a single parameter varied at a time. Each data point is the mean sifted key rate over 1000 Monte Carlo runs; shaded bands show the sample spread. Four sweeps that isolate the dominant geometric and hardware effects are discussed here; the remaining seven are reproduced in Appendix A.

Distance. Figure 5 sweeps Alice–Bob separation from 1–100 km at fixed attenuation.

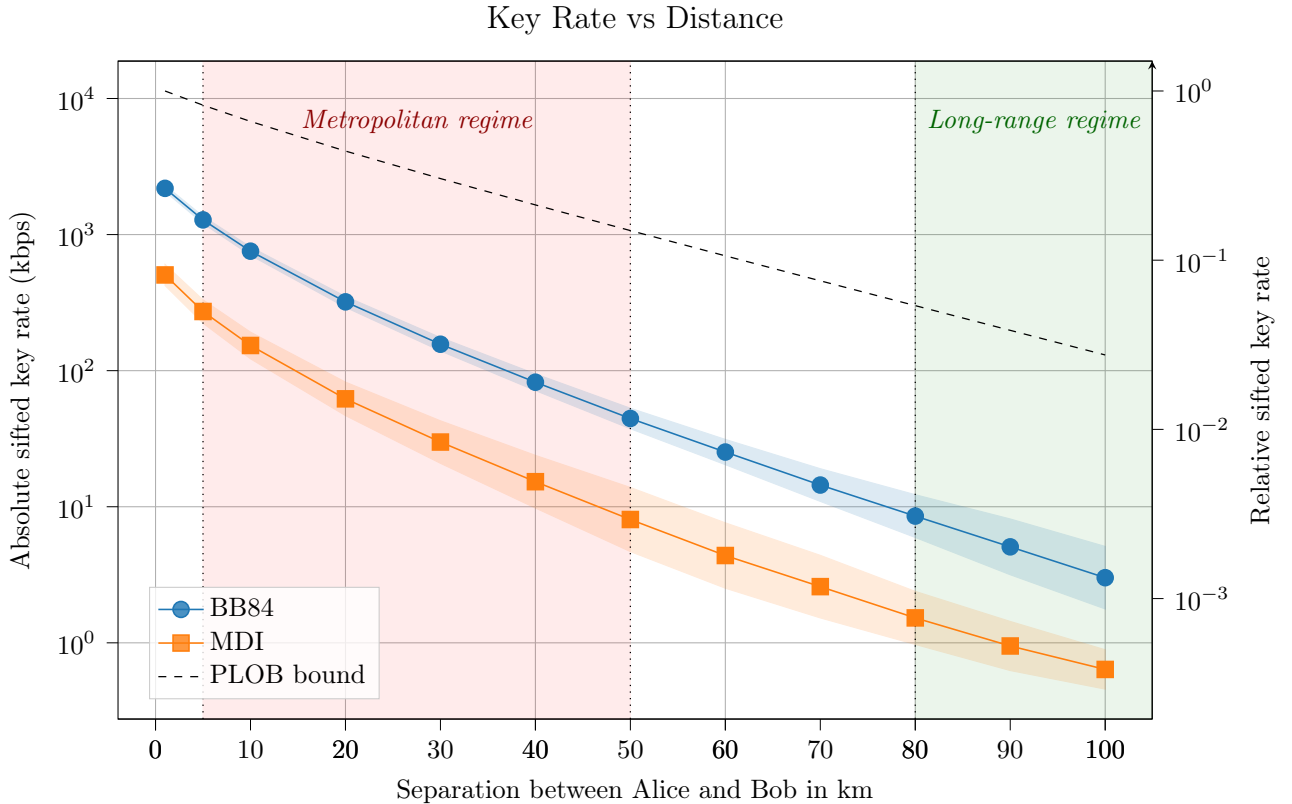


Figure 5: Absolute and relative sifted key rate versus Alice–Bob separation. The PLOB bound is overlaid on the left axis. Shaded vertical bands mark the metropolitan (5–50 km) and long-range (> 80 km) regimes.

The BB84/MDI ratio stays close to a factor of five throughout: $4.3\text{--}5.5\times$ over the full 1–100 km range, varying by no more than $\pm 15\%$ even as key rate itself falls by nearly three orders of magnitude. Both protocols fall increasingly below the PLOB bound with distance: $5.2\times/22.5\times$ at 1 km, widening to $43\times/204\times$ by 100 km for BB84 and MDI, respectively. This provides the practical motivation for the range-extending architectures discussed in

Section 5.1 and revisited in Section 6. This is an offset, not a divergence in decay rate: BB84’s key rate scales linearly in η (Section 5.1), the same high-loss regime the PLOB bound itself approaches ($R_{\text{PLOB}} \approx \eta \log_2 e$, Equation 1), so on a logarithmic axis the BB84 curve runs near-parallel to PLOB at long distance. MDI’s roll-off matches BB84’s rate almost exactly (2.86 vs. 2.90 decades over 1–100 km), since both protocols use the same length of fibre and so link loss scales identically.

Detector- and receiver-side parameters. Figure 6 sweeps detector efficiency η_d from 1.0 down to 0.10, shaded to mark the efficiency bands typical of SNSPD (0.80–0.95, IDQ ID281) and SPAD (0.10–0.25, IDQ ID230) technology, and Figure 7 sweeps receiver node/connector loss from 0–6 dB.

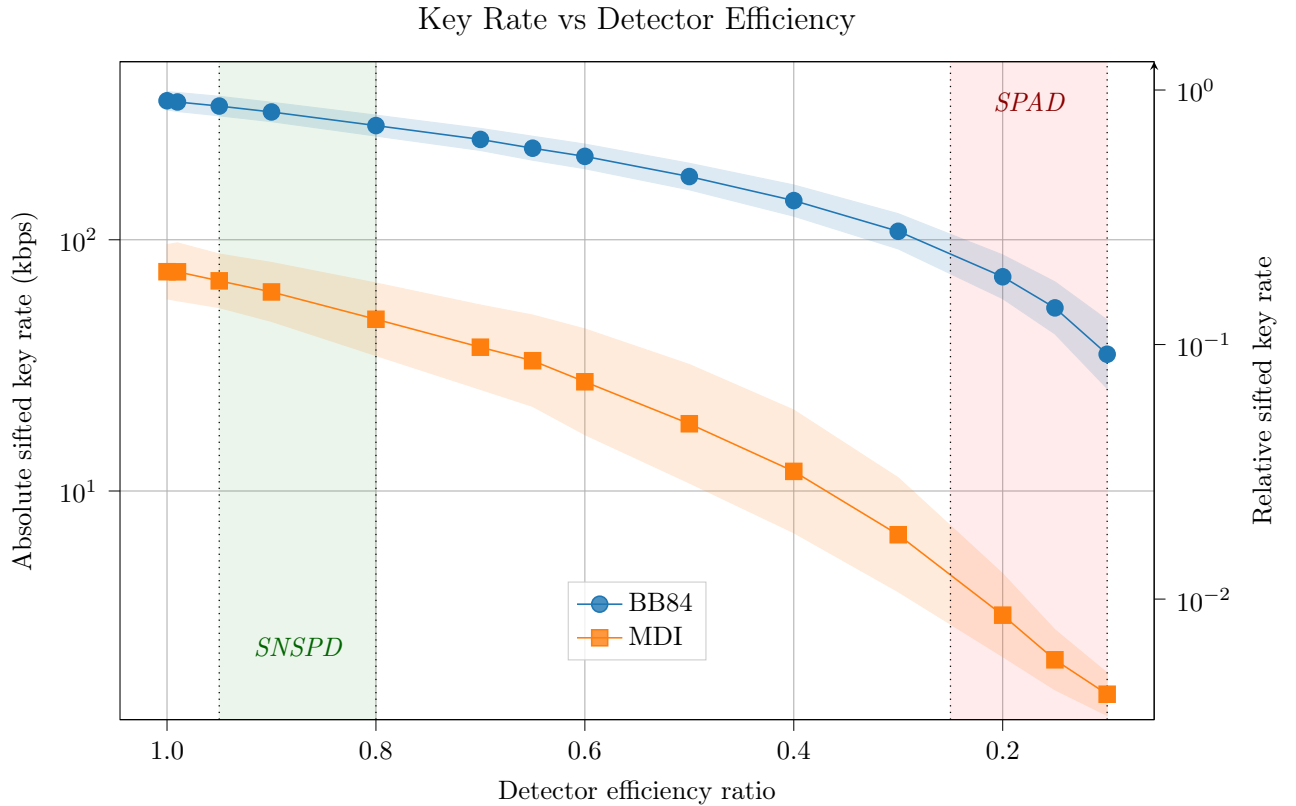


Figure 6: Absolute and relative sifted key rate versus detector efficiency η_d , at fixed 20 km separation and 0.18 dB/km attenuation. Shaded bands mark typical SNSPD (IDQ ID281) and InGaAs SPAD (IDQ ID230) operating ranges.

Unlike distance, the BB84/MDI ratio is not stable under these sweeps: it widens from $4.8\times$ to $25.2\times$ across the bulk of the detector-efficiency range ($\eta_d = 1.0 \rightarrow 0.15$); and from $3.3\times$ to $13.1\times$ under receiver loss (0–6 dB). Both parameters act *per arm*: detector efficiency and receiver loss enter MDI’s coincidence probability twice, quadratically, but BB84’s

detection probability only once (Section 5.1), so any degradation in either hurts MDI disproportionately: detector choice (SNSPD vs. SPAD) therefore matters far more for MDI deployments than for BB84 ones.

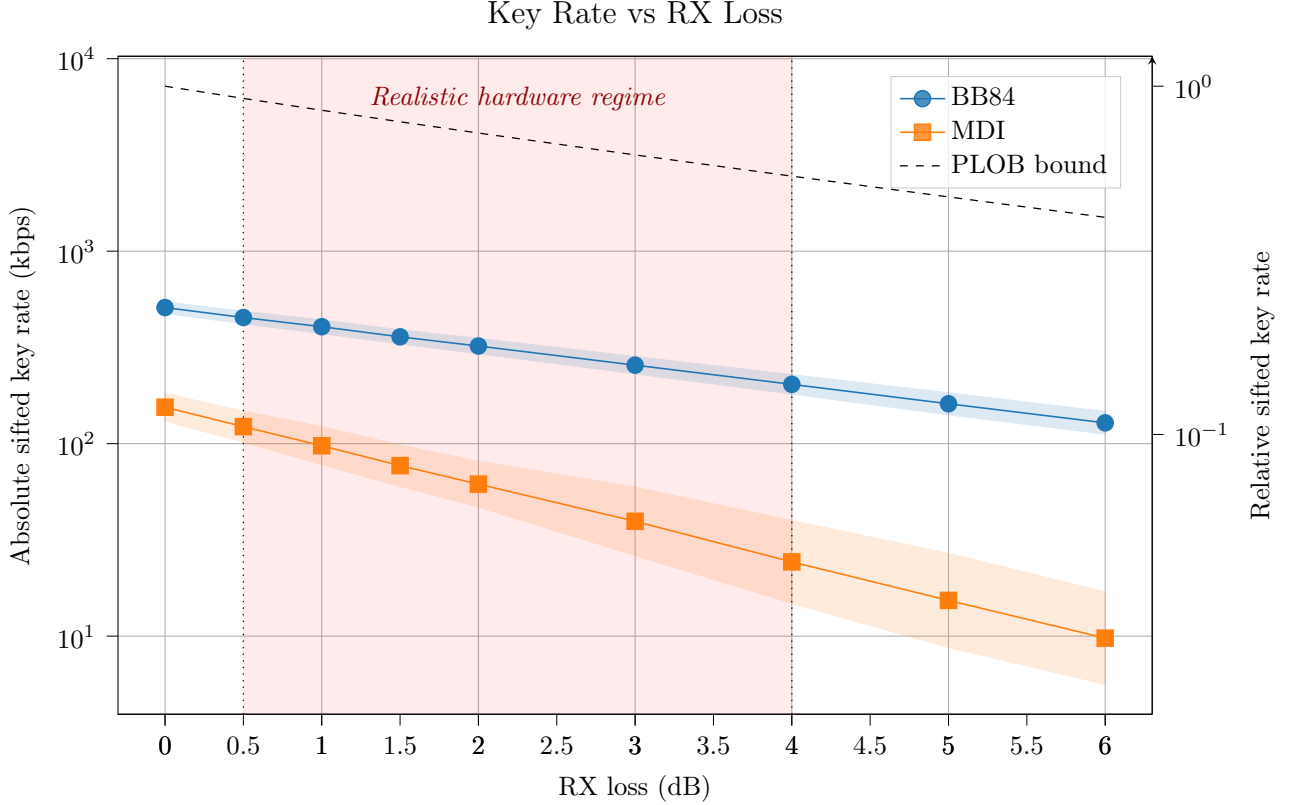


Figure 7: Absolute and relative sifted key rate versus receiver node/connector loss, at fixed 20 km separation and 0.18 dB/km attenuation. The PLOB bound is overlaid on the left axis; the shaded band marks the realistic hardware regime (0.5–4.0 dB).

Relay placement. Figure 8 sweeps Charlie’s position as a fraction of a fixed 20 km Alice–Bob distance.

MDI key rate peaks at the symmetric midpoint (61.7 kbps at 0.5) and falls by over a third towards either end (39.0 kbps at 0.1; 39.2 kbps at 0.9), confirming that Charlie sitting at the arm midpoint is rate-optimal. Moving Charlie off-centre widens the BB84/MDI ratio to 8.2–8.3 $\times$ at the sweep extremes, close to a 60% increase over the midpoint ratio of 5.2 $\times$: a clean illustration that relay geometry, and not only raw hardware quality, materially affects the rate trade-off of choosing MDI over BB84.

Together, these four sweeps indicate that within the metropolitan and realistic hardware regimes studied here, the BB84/MDI key rate gap is set primarily by a roughly constant structural factor of $\sim 5\times$ arising from the two-photon BSM requirement, and that per-arm

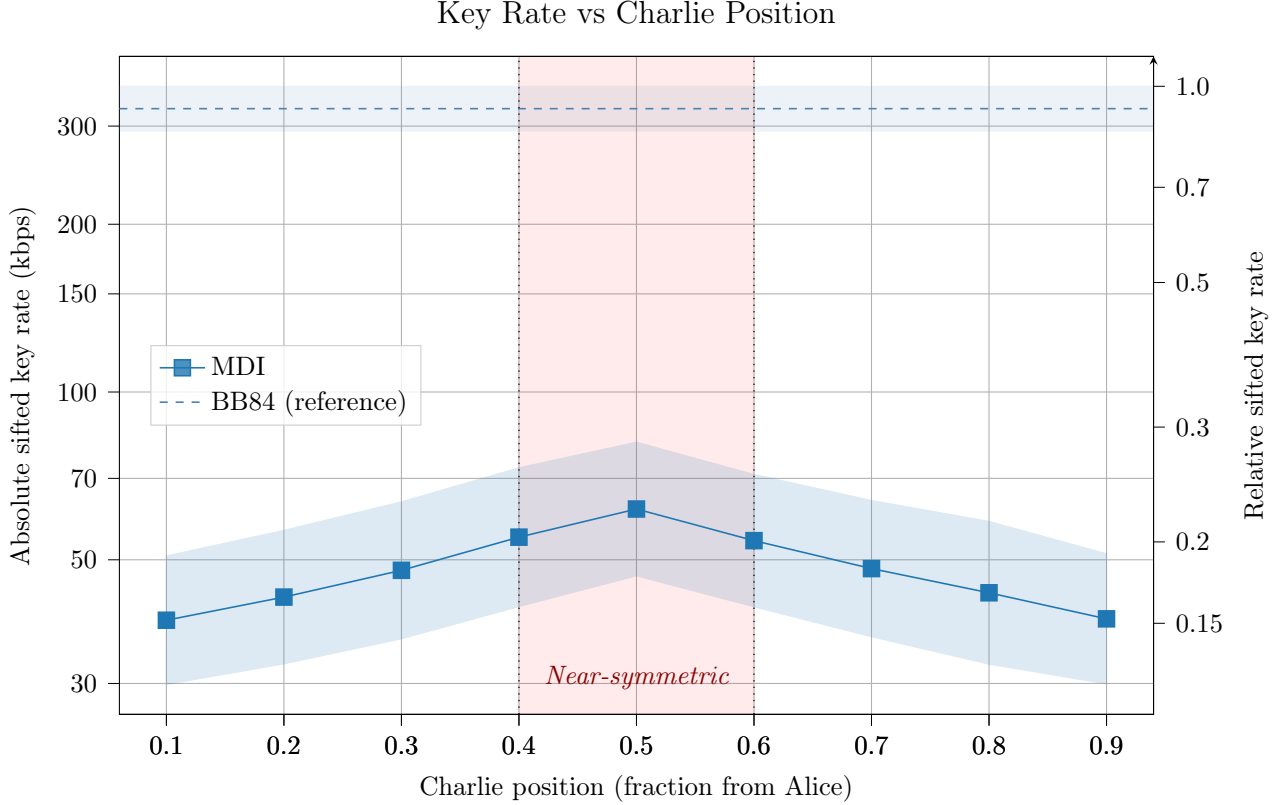


Figure 8: MDI key rate versus Charlie’s position as a fraction of the fixed 20 km Alice–Bob distance, with BB84 plotted as a distance-independent reference. The shaded band marks the near-symmetric regime.

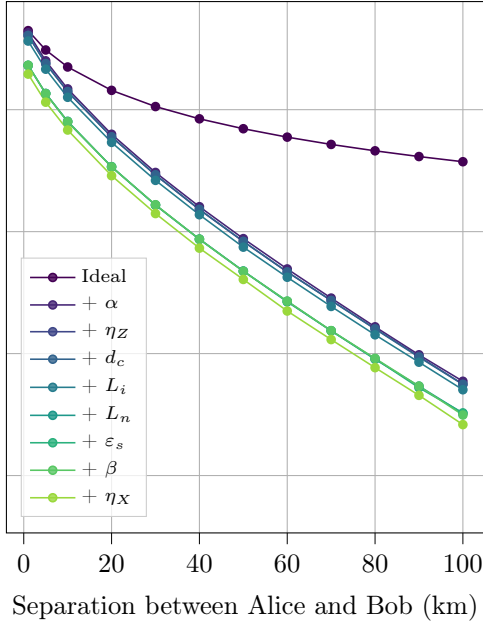
hardware parameters – detector efficiency and receiver loss – are what is most likely to widen this gap substantially in practice. We test this conclusion directly with three further views of the same operating point: the cumulative hardware configurations of Table 2 each distance-swept, a fixed-distance ranking of all nine parameters by relative impact, and the isolated per-parameter distance sweeps that connect the two.

Sample spread. A further pattern, visible in the shaded bands of every sweep above rather than in any single one, is that MDI’s sample spread is consistently wider than BB84’s relative to its own mean. Measuring each shaded band’s half-width as a fraction of its midpoint value and averaging across a sweep, MDI runs $1.7\text{--}3.3\times$ wider than BB84 at the same points: distance (0.71 vs. 0.41), receiver loss (0.65 vs. 0.20), detector efficiency (0.69 vs. 0.28), and Charlie position (0.54 vs. 0.18). This is not an artefact of any one parameter and has two compounding causes. First, MDI’s key generation is conditioned on a coincidence event at the BSM requiring both Alice’s and Bob’s photons to survive their respective arms and arrive indistinguishably within the coincidence window, so an unlucky joint outcome on either arm alone is enough to suppress a trial’s yield. Second, because MDI’s absolute key

rate is intrinsically lower than BB84’s at every operating point studied here, the same Monte Carlo budget yields fewer successful key-generation events per point, so data points carry proportionally more sampling noise by default. The practical consequence is that MDI’s key rate is more variable than BB84’s under otherwise identical conditions, a further, statistical, cost against which its device-independence guarantee must be weighed (Section 5.1).

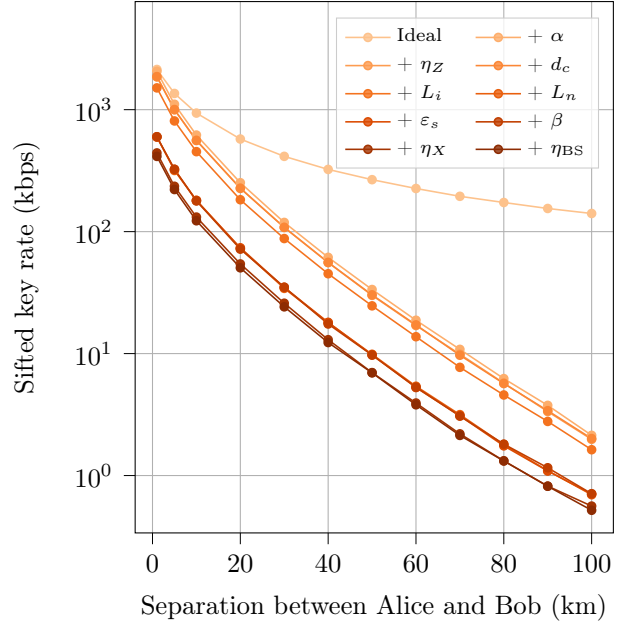
Cumulative hardware configurations. Figure 9 sweeps distance for each of the ten cumulative hardware configurations of Table 2 in turn.

BB84: Cumulative Modelling Layers



(a) BB84

MDI: Cumulative Modelling Layers



(b) MDI

Figure 9: Sifted key rate versus Alice–Bob separation under each cumulative hardware configuration of Table 2. Each configuration includes all impairments of the previous one plus the parameter named in its legend entry. Sample spread is omitted here for clarity (see full point-to-point sweeps for error bands).

Both protocols show the same qualitative shape: a large step when fibre attenuation is first added, followed by a sequence of comparatively small further steps as the remaining impairments accumulate. For BB84 the step from the ideal baseline to fibre attenuation alone is barely visible at 1 km – 4447.7 kbps falls by 3.9% to 4274.8 kbps – but dominates entirely by 100 km, with the same step reducing the rate by 98.4%. MDI shows an identical pattern: 3.4% at 1 km widening to 98.5% at 100 km. The fractional impact of fibre attenuation is itself a function of distance, negligible at short range and overwhelming at long range, consistent with it being the sole impairment whose loss probability scales with L directly (Section 3.1.2) rather than acting as a fixed multiplicative factor. Against this backdrop, node loss stands out among the remaining impairments as the one further step large enough to see clearly

even at short range: for BB84 adding node loss at 1 km induces a 36.8% key rate reduction; for MDI, a 60.4% reduction. Both figures are consistent with the sensitivity ranking below, and both closely match the isolated node-loss impact reported next.

Parameter sensitivity ranking. Figure 10 ranks all ten hardware parameters directly by their individual contribution to key-rate loss, at a fixed 25 km operating point (ideal key rate 1223.40 kbps for BB84, 481.63 kbps for MDI), with all other parameters held ideal. Detector efficiency contributes two bars, one for the SPAD (IDQ ID230, $\eta_d = 0.20$) and another for the SNSPD (IDQ ID281, $\eta_d = 0.90$).

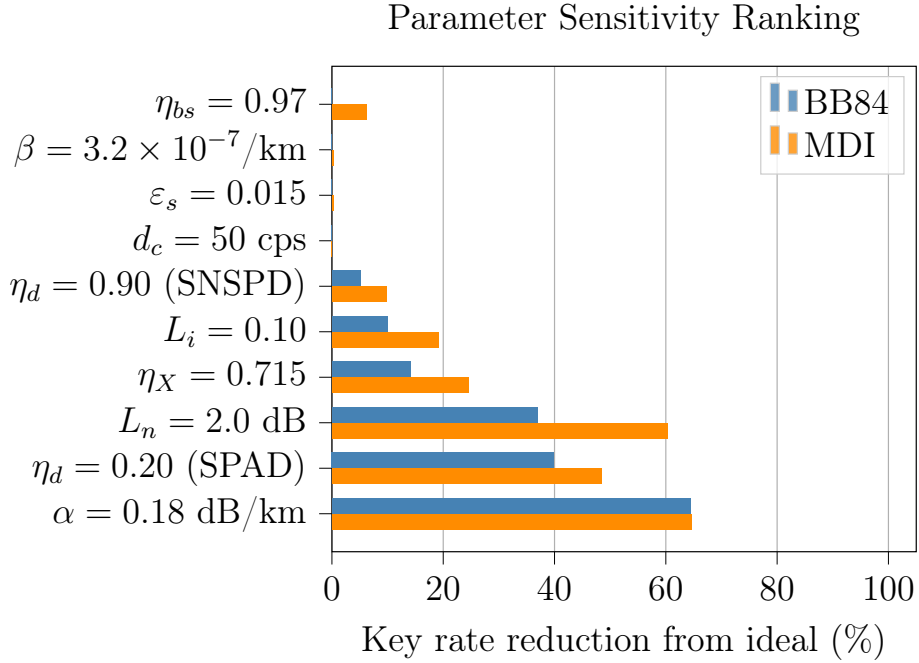


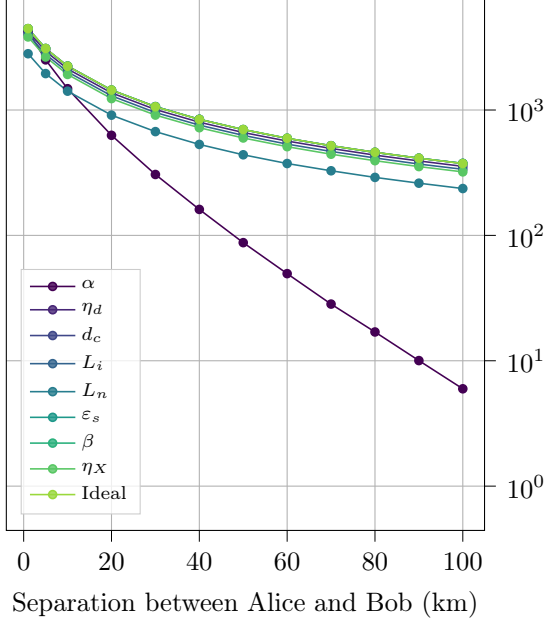
Figure 10: Percentage key-rate reduction from the ideal baseline at 25 km separation, for each parameter set individually to its realistic value (Table 2), ranked by impact on BB84 key rate.

Fibre attenuation dominates both protocols almost identically at this fixed distance (64.4% BB84, 64.7% MDI), the two curves not yet having diverged the way they do by 100 km in Figure 9. The SPAD detector-efficiency bar is the second-largest contributor for BB84 at a 39.9% reduction, and impedes MDI by 48.4%; the SNSPD bar, evaluated at the same operating point, produces 5.1% and 9.8% reductions, respectively. Every other parameter reproduces the per-arm asymmetry established in Section 4.1: node loss reduces BB84 by 36.9% but MDI by 60.3%; basis bias by 14.1% versus 24.5%; TX insertion loss by 10.0% versus 19.2% – each roughly 1.6–2× more damaging to MDI than to BB84. Beam-splitter efficiency, applicable to MDI alone, contributes a further 6.3%. Dark count rate, source error rate, and fibre dephasing are negligible for both protocols at this operating point (under 1% in every case), matching the "noise-like parameters" grouping of the appendix sweeps

(Appendix A).

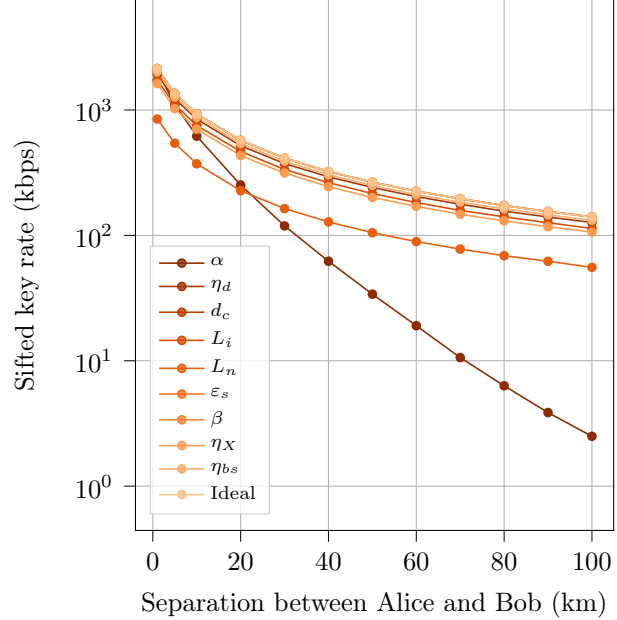
Isolated parameter impact. Figure 11 extends the fixed-distance ranking above across the full 1–100 km range, setting exactly one parameter to its value at a time against an otherwise fully ideal baseline.

BB84: Isolated Parameter Impact



(a) BB84

MDI: Isolated Parameter Impact



(b) MDI

Figure 11: Sifted key rate versus Alice–Bob separation with exactly one hardware parameter set to its fully realistic value (Table 2), against an ideal (baseline) reference. Curves show the Monte Carlo mean.

The two parameters singled out above behave structurally differently across distance. Fibre loss tracks the ideal baseline closely at short range, pulling away with distance (BB84 reduction widens from 4.3% at 1 km to 98.4% at 100 km, MDI from 3.7% to 98.2%). Node loss instead reduces key rate by a *near-constant fraction at every distance*: 36.9% at 1 km versus 36.8% at 100 km for BB84, and 60.3% at both distances for MDI, matching the 25 km sensitivity figures to within Monte Carlo noise. Node loss, basis bias, TX insertion loss, and (for MDI) beam-splitter efficiency therefore constitute a qualitatively distinct category of impairment from fibre attenuation: fixed-fraction, per-arm penalties eroding key rate independent of separation and consistently favouring BB84, rather than range-dependent losses, equivalent between protocols, only becoming significant as a link extends.

4.2 Network Performance

The user-count and relay-count sweeps of Section 3.3 are reported together here, since both characterise network-level key rate as a function of the same underlying variables (N and K) rather than testing independent hypotheses.

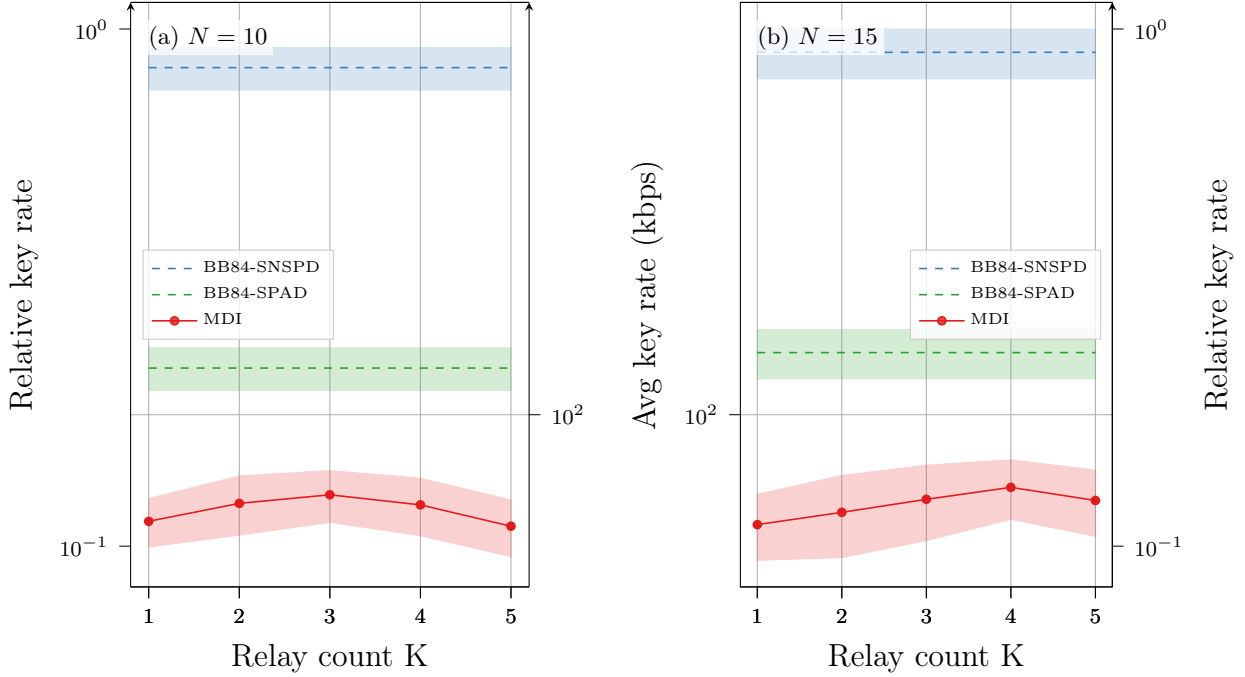


Figure 12: Mean key rate versus relay count K for $N = 10$ (left) and $N = 15$ (right) users in a 25×25 km service area, each averaged over 5 independent seeds; shaded bands show the spread across seeds. BB84 is relay-independent (dashed baselines) and indicated for both SNSPD and SPAD detectors. Both panels share the same y -axis scale on both axes to support direct visual comparison.

MDI’s mean key rate peaks at $K = 3$ for $N = 10$ (67.0 kbps, up 14.2% from $K = 1$, recovering 11.8% of the BB84 SNSPD baseline) and at $K = 4$ for $N = 15$ (69.5 kbps, up 20.5%, recovering 11.3%), falling off past the peak at both user counts (Figure 12). This non-monotonic shape reflects two effects of opposite sign (Section 5.1): more relays shorten the mean user-to-relay spoke distance, improving the coincidence probability, but also fragment the network into smaller catchments, raising the chance a user pair must route via an inter-relay backbone hop. The peak arriving one relay earlier at the smaller user count implies that the rate-optimal K scales like $\sqrt{N}$, discussed in more detail in Section 6.3, though the recovered fraction of the BB84 baseline is practically unchanged. Table 4’s per-seed breakdown for $N = 15$ confirms the mean’s $K = 4$ peak is not an averaging artefact: four of five seeds peak there outright, and the fifth is within Monte Carlo noise of doing so.

Total fibre length, by contrast, is minimised around $K = 2$ – 3 at both user counts, not at

Seed	Metric	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$
87100	Key rate (kbps)	48.5	47.2	54.4	62.9	56.3
	Fibre length (km)	145.8	123.8	115.8	129.0	143.3
29252	Key rate (kbps)	64.8	74.5	80.7	78.9	76.3
	Fibre length (km)	121.0	87.0	88.1	105.7	115.8
44358	Key rate (kbps)	51.2	50.5	51.7	55.8	55.8
	Fibre length (km)	137.9	111.3	112.0	126.0	135.2
91457	Key rate (kbps)	51.2	56.9	61.4	66.4	57.1
	Fibre length (km)	134.5	104.7	101.9	118.6	131.0
39964	Key rate (kbps)	72.7	77.6	79.2	83.6	79.9
	Fibre length (km)	112.3	84.6	89.0	98.2	107.3
Mean	Key rate (kbps)	57.7	61.3	65.5	69.5	65.1
	Fibre length (km)	130.3	102.3	101.3	115.5	126.5

Table 4: MDI mean key rate and total fibre length (Equation 7, straight-line Euclidean distance without stochastic tortuosity) versus relay count K , per-seed, for $N = 15$. Bold marks each row’s optimum.

the key-rate optimum (Table 4): the backbone term grows quadratically in K (Equation 7) while the spoke term can only fall so far, so past that point each further relay costs more backbone fibre than it saves on spokes, even as it continues to help key rate by shortening the worst-case spoke distance. The fibre-cost-optimal and key-rate-optimal relay counts are therefore different operating points: pushing from the fibre-optimal K to the key-rate-optimal one costs roughly 13–14% more fibre at $N = 15$.

A cheaper SPAD detector ($\eta_d = 0.20$) does not close the gap with SNSPD-MDI at either user count: the SPAD-BB84 baseline (126.3 kbps at $N = 10$, 136.5 kbps at $N = 15$) sits roughly 1.9–2.0 $\times$ above even the SNSPD-MDI peak, with no overlap between the two distributions at either N . Any cost comparison between the two protocols (Section 4.3) must therefore be read as a comparison at vastly unequal key rates, not as a search for a SNSPD-MDI deployment matching SPAD-BB84’s throughput.

4.3 Cost

Using the component counts of Table 3 (Section 3.2), the total CapEx for each protocol is

$$C_{\text{BB84}} = N c_s + 2N c_d + \frac{N(N-1)}{2} \bar{d}_b c_f, \quad (10)$$

$$C_{\text{MDI}} = N c_s + 4K c_d + \left(N \bar{d}_u + \frac{K(K-1)}{2} \bar{d}_r \right) c_f, \quad (11)$$

where $\bar{d}_b$ is the mean user-pair distance in the BB84 mesh, $\bar{d}_u$ the mean user-to-relay distance, and $\bar{d}_r$ the mean relay-to-relay backbone link length. Passive relay hardware (HOM beam

splitters, polarising beam splitters, optical switches) contributes small additional fixed cost per relay and is omitted here for simplicity; a deployed metropolitan MDI-QKD network using exactly this apparatus is reported in [16].

The fibre term dominates the cost difference at large N , since it scales as $\mathcal{O}(N^2)$ for BB84 against $\mathcal{O}(N)$ for MDI at fixed K . Thus, a crossover user count N^* past which MDI is cheaper always exists. Setting $C_{\text{BB84}} = C_{\text{MDI}}$ and solving for N gives N^* as a function of the ratio c_d/c_f , relay count K , and network geometry. Evaluating N^* explicitly requires relations between the distance term $\bar{d}_b$ in Equation 10 and the terms $\bar{d}_u$ and $\bar{d}_r$ in Equation 11.

Neither is a free parameter in practice: both follow from standard results in geometric probability, assuming relay placement is roughly uniformly across the service area (Section 3.1.4). For m points scattered independently and uniformly over a square of side L , the mean pairwise distance converges to

$$|\overline{PQ}| = \frac{2 + \sqrt{2} + 5 \operatorname{arcsinh}(1)}{15} L \approx 0.5214 L, \quad (12)$$

independent of m [42] (what do you mean convergence and independence here?). Setting $m = N$ is what already makes $\bar{d}_b$ itself independent of N in Equation 10; setting $m = K$ instead applies the same fact to the K relay positions, giving

$$\bar{d}_r \approx \bar{d}_b. \quad (13)$$

Spoke distance is a different quantity. K relays partitioning the service area into roughly equal-area catchments gives each catchment area $\approx L^2/K$, i.e. linear (side) dimension $\ell \approx L/\sqrt{K}$; each user's assigned relay sits near that catchment's geometric median, close to its centre. The mean distance from a point drawn uniformly at random within a square of side ℓ to that square's own centre is a second, distinct geometric-probability constant [42]:

$$|\overline{PO}| = \frac{\sqrt{2} + \ln(1 + \sqrt{2})}{6} \ell \approx 0.3826 \ell. \quad (14)$$

Substituting $\ell \approx L/\sqrt{K}$ gives $\bar{d}_u \approx 0.3826 L/\sqrt{K}$; eliminating the shared, unknown L using Equation 12 in its place yields the working approximation used below:

$$\bar{d}_u \approx \frac{0.3826}{0.5214} \cdot \frac{\bar{d}_{\text{BB84}}}{\sqrt{K}} \approx 0.73 \frac{\bar{d}_{\text{BB84}}}{\sqrt{K}}. \quad (15)$$

Both approximations are taken in the regime $N \gg K$: each relay serving many users, the regime characteristic of realistic large-scale MDI-QKD deployment via wavelength- or time-division multiplexing. In this regime a catchment's users fill its area densely enough for the

uniform, independent scattering assumed by Equations 12 and 14 to hold.

For the analysis that follows, we additionally assume multi-channel detector modules suitable for every node's needs. The IDQ ID281 SNSPD system's 16 detection channels evidences that just one module can be necessary for a single node's detector channel requirement. Under this assumption BB84's detector cost becomes $N c_d$ and MDI's becomes $K c_d$, assuming symmetric cost of detector systems. We write c_F for $c_f d$, the average full-mesh link cost.

Substituting Equations 13 and 15, together with the multi-channel detector assumption above, into Equations 10–11 (writing $d \equiv \bar{d}_{\text{BB84}}$) gives a closed form for the cost difference, with the $N c_s$ source term cancelling identically:

$$\Delta C \equiv C_{\text{BB84}} - C_{\text{MDI}} \approx c_d(N - K) + c_F \left[\frac{N(N - 1)}{2} - 0.73 \frac{N}{\sqrt{K}} - \frac{K(K - 1)}{2} \right]. \quad (16)$$

Expanding as a polynomial in N ,

$$\Delta C \approx \frac{1}{2} c_F N^2 + \left(c_d - c_F \left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right) \right) N - \left(c_d K + \frac{1}{2} c_F K(K - 1) \right), \quad (17)$$

with no further approximation. The linear coefficient's bracketed factor $\frac{1}{2} + \frac{0.73}{\sqrt{K}}$ decreases monotonically in K towards $\frac{1}{2}$ as $K \rightarrow \infty$; the coefficient itself therefore exceeds $c_d - c_F$ for all $K \geq 3$, and falls only marginally below it at $K = 2$.

Equation 17 is quadratic in N and solves in closed form. Writing $\rho \equiv c_d/c_F$ for the dimensionless ratio of single detector-module cost to average single full-mesh link cost,

$$N^*(\rho, K) = - \left(\rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}} \right) + \sqrt{\left(\rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}} \right)^2 + 2\rho K + K(K - 1)}. \quad (18)$$

$N^*(\rho, K)$ is monotonically decreasing in ρ for K fixed: differentiating Equation 18 and simplifying, so its value at $\rho = 0$ is a hard ceiling:

$$N^*(0, K) = \left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right) + \sqrt{\left(\frac{1}{2} + \frac{0.73}{\sqrt{K}} \right)^2 + K(K - 1)}. \quad (19)$$

For $K \geq 3$, the first term is less than 1, and the square root is $\leq K$. N^* evaluates to 2.76 at $K = 2$, and so for each fixed $K \geq 2$, $N^* \leq K + 1^1$. Aside from the fact that the simulation data points to MDI key rate peaking for $K = \sqrt{N}$, economic deployments of MDI-QKD would target far fewer relays than users (i.e. $K \ll N$). Thus, for every realistic (N, K) combination, $N > N^*(\rho, K)$ for *every* $\rho \geq 0$, so **MDI is the cheaper protocol to deploy at every operating point, regardless of detector price** conditioned only on symmetric

¹ $N^*(0, 1) = 2.46$, the *smallest* crossover value at any relay count. Even so, only $N = 3$ users are needed for MDI to be strictly cheaper than BB84 at $K = 1$, regardless of detector price.

price for detector systems across protocols.

If $K = N$, then $\Delta C(K) = -0.73 c_F \sqrt{K}$, negative for every $K > 0$ and independent of c_d (the detector term vanishes identically). BB84 thus remains the cheaper protocol if all users are also relays, regardless of detector price, and a crossing never occurs. $K = N$, however, is precisely the deployment no operator would choose: a relay serving a single user shares no fibre and provides none of the aggregation MDI's hub topology exists to offer. The headline result above therefore holds across the entire range of *sensible* deployments, and fails only where relaying itself would already be pointless.

The K floor is exactly attained in the limit. Writing $a \equiv \rho - \frac{1}{2} - \frac{0.73}{\sqrt{K}}$, the identity $\sqrt{a^2 + c} - a = c/(\sqrt{a^2 + c} + a)$ applied to Equation 18 (with $c = 2\rho K + K(K-1)$) gives

$$N^*(\rho, K) = \frac{2\rho K + K(K-1)}{\sqrt{a^2 + 2\rho K + K(K-1)} + a}. \quad (20)$$

The numerator grows as $2K\rho$ and the denominator as 2ρ , so $N^*(\rho, K) \rightarrow K$ as $\rho \rightarrow \infty$, with $N^*(\rho, K) - K \sim 0.73\sqrt{K}/\rho$ at next order. Combined with the $\rho = 0$ ceiling above, N^* is confined to the open interval $(K, K+1)$: less than one user's width, regardless of how cheap or expensive detectors become.

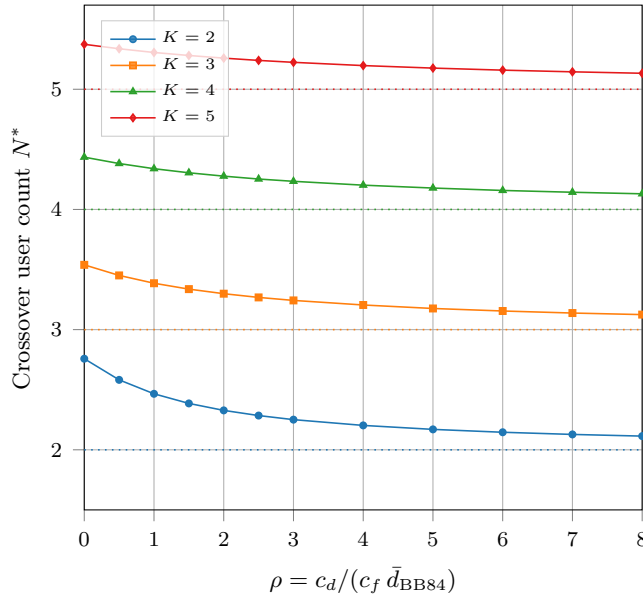


Figure 13: Crossover user count N^* against the dimensionless price ratio $\rho = c_d/(c_f \bar{d}_{\text{BB84}})$, from the closed-form solution of Equation 18, for $K = 2-5$; dotted lines mark each curve's $N^* \rightarrow K$ asymptote, approached from above.

Figure 13 makes this visible directly: each curve falls from its $\rho = 0$ ceiling towards its floor (dotted) as $\rho \rightarrow \infty$.

Marginal cost of network growth. The comparison above treats N as fixed; a provider

deciding whether to extend an existing network by one more user instead faces a marginal-cost question, holding K fixed. Differencing Equations 10–11 (with the multi-channel detector assumption above) between N and $N + 1$ gives the cost of the $(N+1)$ -th user under each protocol:

$$\Delta C_{\text{BB84}}^{\text{marg}}(N) = c_s + c_d + c_F N, \quad \Delta C_{\text{MDI}}^{\text{marg}}(N) = c_s + c_f \bar{d}_u \approx c_s + 0.73 \frac{c_F}{\sqrt{K}}. \quad (21)$$

BB84’s marginal cost grows linearly with the size of the network the new user joins: the new source and detector module cost the same regardless of N , but the number of fibre links required grows with N , since new users connect to all existing ones. MDI’s marginal cost is instead constant in N : the new user needs only one spoke link, of mean length $\bar{d}_u$, to its existing relay, and shares that relay’s already-provisioned detector module regardless of how many other users it already serves. The difference between the two,

$$\Delta C_{\text{BB84}}^{\text{marg}}(N) - \Delta C_{\text{MDI}}^{\text{marg}}(N) = c_d + c_F \left(N - \frac{0.73}{\sqrt{K}} \right), \quad (22)$$

is strictly positive for every $N \geq 1$ and $K \geq 2$ regardless of c_d and c_f : **adding a user to an existing MDI network is cheaper than adding one to a BB84 mesh of the same size**, at every step, unconditionally on detector price. This is the mechanism behind the tight N^* ceiling derived above: BB84 is only ever cheaper because of a smaller *initial* outlay, and every user added costs strictly more, so that initial head start is quickly lost. In this light N^* marks not a lasting cost advantage but the point at which BB84’s one-off discount is fully repaid.

One caveat attaches to the multi-channel assumption above: in reality single-photon detector systems are not as configurable as our analysis suggests. Custom detection channel-count devices do not yet exist, and even if they did we make the assumption that a four-channel detector costs the same as a two-channel detector. Further, as seen in Figure 12, BB84 does not require the same high detection efficiencies that MDI does for the same key rates (and it still outperforms), reducing the required cost for BB84 detectors - but as outlined initially, fibre scaling means that there will still always be a crossing point N^* for large enough networks.

4.4 Deployment Recommendations

Cost and key rate are not the only axes that differentiate the protocols in this work, and for many considering MDI, they are not even the primary one: MDI-QKD’s principal selling point is its security guarantee, not its throughput. It is useful to ask directly what a deployment

requires the customer to trust: the photon source, the detectors, any relay hardware, and the people operating that hardware. This is what varies between architectures and what a customer’s threat model actually constrains. One component never varies: the quantum channel itself (the fibre, and by extension any eavesdropper on it) is assumed fully adversarial at every level - that is the premise QKD exists to satisfy, not a design choice. Customer trust requirements give a strict hierarchy, each assignable a set of protocols:

- **Level 0: Trust nothing.** Not the source, not the detectors, not any relay, not any operator. This is the ceiling fully Device-Independent QKD aims for, first proposed with a security proof against collective attacks by [43] and recently demonstrated experimentally [44]; it is not implemented in this work.
- **Level 1: Trust in sources.** Detectors, relay hardware, and the relay operator are all permitted to be fully adversarial: Charlie’s Bell-state measurement discloses only the joint correlation between Alice’s and Bob’s photons (Section 5.1). The source is still trusted, though multi-photon emission and Trojan-horse side channels are residual vulnerabilities that decoy-state protocols [10, 29] address but the ideal sources used in this work do not model directly (Section 5.3). No substitute exists at this level: a customer whose threat model stops at Level 1 has no BB84 alternative.
- **Level 2: Trust in sources and detectors.** Adds detector trust to Level 1’s source trust, in exchange for removing the relay from the picture entirely: with no third party involved, there is nothing further to trust, but the link cannot extend beyond direct point-to-point range. Detector trust is precisely the assumption BB84 requires and MDI was invented to remove (Section 5.1), so this level is only appropriate when both endpoints already fall within a single security perimeter.
- **Level 3: Trust everything.** Adds trust in relay hardware *and* its operator: a compromised relay node, or a dishonest operator, recovers the full end-to-end key for any pair routed through it (Section 5.1). In exchange, range extends via relay hops at BB84’s rate and cost, without MDI’s two-photon coincidence penalty. Appropriate only when the relay is under the customer’s own physical and operational control, or is otherwise contractually and regulatorily accountable, as in deployed trusted-node metropolitan QKD networks such as the 46-node Hefei network [45].

Each level’s trusted set strictly contains the one below it, so a customer’s threat model fixes a level first, and only *then* does the cost/rate trade-off below apply, choosing among whichever architectures satisfy it. A customer requiring Level 1 is not, at that point, choosing between BB84 and MDI at all: MDI is the only option that meets the requirement.

Protocol selection recommendations follow from combining the simulation key rate results with the algebraic cost model. The relay sweep provides mean key rate $\bar{R}(K)$ at fixed N and area; Equations 10–11 express cost as a function of the same variables and the free parameters c_s, c_d, c_f .

For a provider with a minimum key rate requirement $R_{\min}$, the relay sweep identifies the minimum relay count K^* satisfying $\bar{R}(K^*) \geq R_{\min}$. The corresponding MDI cost $C_{\text{MDI}}(N, K^*, c_s, c_d, c_f)$ can then be compared to the BB84 cost $C_{\text{BB84}}(N, c_s, c_d, c_f)$ symbolically, revealing the conditions on c_d/c_f under which MDI's security advantage is obtained within the same budget as BB84. The crossover user count N^* derived from the cost formulae provides a scale threshold independent of absolute component prices.

5 Discussion

The results presented in Section 4 support a consistent picture of the trade-off space between BB84 and MDI-QKD. We discuss the key findings, their physical interpretation, the limitations of the methodology, and their implications for protocol selection.

5.1 The Security–Rate Trade-off

MDI-QKD’s elimination of detector side-channel attacks comes at a key rate penalty rooted in its physical architecture. Whereas BB84 requires one photon to traverse the full link and be detected, MDI requires two photons — one from each party — to arrive at Charlie’s relay simultaneously and produce a coincidence event. Loss on each arm compounds multiplicatively: if the single-arm transmittance is η_{arm} , the BSM detection probability scales as $\eta_{\text{arm}}^2 \eta_{\text{BS}} \eta_d^2$, compared to $\eta_{\text{link}} \eta_d$ for BB84. At metropolitan distances and typical detector efficiencies, this effectively squares the channel transmittance, consistent with the order-of-magnitude rate gap between the two protocols in the bipartite setting. This is borne out empirically by the point-to-point sweeps in Section 4.1: the BB84/MDI ratio stays close to a constant $\sim 5\times$ under channel-side sweeps (distance, attenuation) but widens to over $25\times$ under a detector efficiency sweep (Figure 6), the signature of a parameter that acts quadratically on MDI’s coincidence probability but only linearly on BB84’s single-photon detection probability. The sensitivity ranking in Figure 10 quantifies this split directly at a fixed 25 km operating point: fibre attenuation degrades both protocols by almost the same fraction (64.4% BB84, 64.7% MDI), while every per-arm parameter — node loss, basis bias, TX insertion loss, detector efficiency, and (MDI-only) beam-splitter efficiency — degrades MDI by roughly $1.6\text{--}2\times$ more than BB84 in relative terms. The cumulative-configuration and isolated-parameter sweeps of Figure 9 and Figure 11 show why the two channel-side and per-arm categories behave so differently with distance: fibre attenuation’s fractional impact grows from a few percent at 1 km to over 98% at 100 km for both protocols, since loss probability scales with path length directly, whereas node loss removes an essentially constant fraction of key rate — around 37% for BB84 and 60% for MDI — at every distance from 1–100 km, since it is a fixed per-arm insertion loss applied once regardless of link length.

At the network level, the rate gap between MDI and BB84 widens relative to the bipartite case. MDI users route photons to a relay rather than directly to their communication partner, incurring a geometric detour for cross-cluster pairs. This routing overhead is absent in BB84’s direct mesh. The relay distance advantage is a function of relay count K and user distribution; the relay sweep quantifies the extent to which increasing K can recover the MDI rate penalty.

At $N = 15$ (Figure 12b), increasing K from 1 to 4 recovers only a small fraction of the bipartite MDI/BB84 gap — from 9.4% to 11.3% of the BB84 rate — and the recovery is non-monotonic, reversing by $K = 5$: shortening the mean spoke distance helps only up to the point where the resulting catchment fragmentation starts forcing more user pairs through an additional backbone hop.

A third trust model is available in the simulation infrastructure: trusted-node BB84 (TBB84) [13], implemented in `trusted_bb84_network.py` and described in Section 3.1.4. TBB84 routes a key between non-adjacent users via one or more relay hops, with each hop running standard BB84 and the resulting per-hop keys classically XOR-combined along the path to produce an end-to-end key. This trust model is fundamentally different from MDI’s. MDI’s Bell-state relay discloses only the joint correlation between Alice’s and Bob’s photons; even a fully adversarial Charlie learns nothing about either party’s individual bit values, so no trust in the relay is required. A TBB84 relay, by contrast, handles plaintext-equivalent key material at every hop it forwards: if a relay node is physically compromised, the attacker recovers the full end-to-end key for any pair routed through it. TBB84’s security therefore rests on the physical security of relay hardware, whereas MDI’s rests on detector integrity — the two protocols defend against different adversaries rather than offering directly comparable guarantees.

Because TBB84’s principal advantage over direct BB84 is range extension, obtained precisely by accepting this relay-trust assumption, it is only exercised in this work’s real-topology experiments (Section 3.1.4), where relay hops span distances that a single link could not otherwise reach. Within the 10×10 km synthetic metropolitan grid used for the core BB84-vs-MDI comparison, direct BB84 links and MDI’s Bell-state relay both already achieve useful key rates without range extension (Section 5, *Metropolitan network scope*), so TBB84 would offer no rate or reach advantage there — only the strictly weaker trust assumption of trusted-node relaying, for no compensating benefit at this scale. TBB84 is therefore excluded from the main relay-count and user-count sweeps and reserved for deployments where relay range-extension is structurally required.

5.2 Cost Model Limitations

The cost model is a first-order CapEx-only algebraic estimate. Several factors are not captured.

Component price data. Manufacturer prices for quantum dot photon sources and SNSPD detectors are not publicly available; academic literature reports a wide range of estimates, and commercial quotes are required for quantitative comparison. The symbolic formulation (Equations 10–11) defers this uncertainty to c_s , c_d , and c_f , but the sensitivity of the crossover

N^* to these parameters is significant: under the one-detector-module-per-node assumption of Section 4.3, a high c_d/c_f ratio (expensive detectors relative to fibre) instead *narrows* BB84’s advantage, since BB84 needs one module per user against MDI’s one per relay and so carries the larger total detector count for realistic $N \gg K$, while a high c_f/c_d ratio widens it by letting BB84’s smaller- N fibre saving matter more relative to detector cost.

Symmetric detector pricing. The model uses a single price c_d for both protocols’ detector modules throughout Section 4.3, treating BB84 and MDI detector costs as identical. In practice they need not be: MDI’s Bell-state measurement requires photon indistinguishability at the relay’s beam splitter, plausibly demanding tighter timing-jitter and inter-arm efficiency matching than BB84’s simpler single-photon detection and commanding a price premium as a result; conversely, the SPAD-BB84 deployments considered in Section 4.2 tolerate much lower-specification, cheaper detectors than the SNSPDs assumed for MDI, giving BB84 its own potential cost advantage the symmetric model cannot capture. Every result in Section 4.3 — the N^* bound and the marginal-cost comparison alike — should therefore be read as an all-else-equal comparison at a shared detector price, not a claim that real per-unit prices are equal; Section 6.3 proposes relaxing this directly. It is also worth noting that a price premium on MDI’s detectors would not, on its own, overturn the performance picture either: Section 4.2 shows BB84 fitted with cheap, worse-performing SPADs still achieves a higher key rate than MDI fitted with premium, best-in-class SNSPDs (126.3–136.5 kbps against a 67.0–69.5 kbps SNSPD-MDI peak, with no overlap between the two distributions at either N tested), so any asymmetric pricing that made MDI’s detectors *more* expensive than BB84’s would be compounding an existing rate disadvantage rather than offsetting one.

Operational expenditure. Cryogenic systems — both SNSPD detectors and QD sources — require continuous cooling to temperatures of order 2–4 K, incurring energy costs that can rival hardware CapEx over a ten-year deployment lifetime [23]. The energetic cost of operating K SNSPD-equipped MDI relay stations may be significant at scale and is not captured by the CapEx-only model.

BB84 fibre scaling at large N . The BB84 direct mesh assumes one independent dark-fibre run per user pair. In practice, WDM or switched architectures would serve many pairs over shared fibre infrastructure at high N , substantially reducing BB84 fibre cost. The $\mathcal{O}(N^2)$ fibre scaling should therefore be interpreted as an upper bound applicable to passive, unmanaged dark-fibre mesh deployments; it is physically exact for that specific case.

5.3 Simulation Model Limitations

Source brightness and insertion loss decoupling. The Quandela Prometheus single-photon source used as the reference product operates at 975 nm and achieves a minimum fibred

brightness of 40%, with frequency conversion required to reach the 1550 nm telecom band. In the simulation, source brightness is absorbed into the effective source repetition rate (`source_freq` = 10^7 Hz, consistent with the 32 MHz raw rate after 40% brightness), and the `init_loss` parameter models separate optical path losses independently. Frequency conversion efficiency, additional conversion-stage noise, and the explicit coupling between source brightness and photon rate are not separately modelled. This is a deliberate simplification; a full model would propagate each loss stage independently and account for Raman background photons introduced by the conversion stage.

Ideal single-photon sources. All simulations assume one photon per pulse with no multi-photon component. WCP laser sources are overwhelmingly more practical and affordable than QD sources, and their multi-photon components require decoy-state analysis [29] to extract secure key rate. The ideal source assumption gives an upper bound on key rate and a lower bound on source cost; both should be re-evaluated for WCP deployments.

Sifted, not secure, key rate. Every key rate reported in this work, including R in Equation 4, is a sifted key rate: the length of the matched-basis raw key divided by simulation time, before error correction or privacy amplification (Section 3.1). Standard secure-key-rate formulas subtract an error-correction leakage term — typically the Shannon binary entropy of the QBER, following Lütkenhaus’s analysis for realistic BB84 receivers [46] or the more general GLLP framework for imperfect devices [47] — and shrink the remainder by privacy amplification to bound Eve’s information; more recent numerical approaches, such as the semidefinite-programming framework of Tan et al. [48], compute this bound directly from a device description without a closed-form leakage formula and extend naturally to untrusted-device protocols like MDI-QKD. None of these corrections is applied here. Because the leakage term is QBER-dependent and QBER differs between BB84 and MDI-QKD at a given operating point (Figure 10), the secure-key-rate gap between the two protocols would not simply rescale the sifted-key-rate gap reported throughout Section 4; applying one of these corrections is a prerequisite before the comparison can be read as one of secure, rather than sifted, throughput.

Asymptotic key rates. Finite-key corrections become significant at low photon count rates and short block lengths, reducing the extractable key compared to the asymptotic rate reported here. At long distances where photon detection events are sparse, the finite-key penalty is expected to be most severe for MDI-QKD, potentially widening the gap with BB84 beyond what the asymptotic analysis suggests.

Relay placement. k -means relay placement minimises mean user-to-relay distance but is not provably optimal for network key rate or cost. ILP-based placement [22] could provide tighter lower bounds on minimum MDI deployment cost, and would be a valuable extension

of this work.

Receiver node loss. The model applies a single lumped node loss $L_n = 2.0$ dB uniformly to both BB84 and MDI-QKD receivers. In reality, the optical components at each receiver are protocol-specific: BB84 uses an EOM and PBS, whilst MDI arms use a HOM beam splitter, PBS, and (for cross-cluster routing) an optical switch. These components carry different insertion losses and are purchased and maintained separately. A rigorous model would assign per-component losses derived from product datasheets, track them independently in the cost model, and apply the correct protocol-specific total at each receiver. This is left as future work; the current lumped approximation introduces a systematic bias in the relative key rate comparison that cannot be quantified without the component-level data. The size of this bias is not negligible: Figure 10 shows node loss alone accounting for a 36.9% key-rate reduction for BB84 versus 60.3% for MDI at 25 km, the single largest per-arm contributor to the BB84/MDI gap identified in this work, so a protocol-specific breakdown of L_n would plausibly shift this margin further rather than narrow it, given MDI’s additional relay-side components.

NetSquid computational cost. The simulations in this work model each photon trial as an independent, identically distributed draw with no channel memory or time-varying network state, a regime NetSquid’s discrete-event engine is not strictly necessary for; it was chosen instead for extensibility to temporally correlated dynamics (detector deadtime, jitter, entanglement distillation) that this work does not yet exploit but that matter if this infrastructure is extended to repeater chains or adaptive protocols. That extensibility is not free: DES wall-clock cost constrained the Monte Carlo and seed budgets feasible within the project timeline and was the binding constraint on the network sizes explored in Section 3.1.4. Appendix D quantifies this cost, the mitigation strategies attempted during this work (adaptive Monte Carlo sampling, Gaussian process surrogate modelling), and the further engineering — batched trial execution, a persistent worker pool, a distributed DES backend — that a commercial-scale successor to this infrastructure would need.

Metropolitan network scope. The core BB84-vs-MDI comparison models networks within a 10×10 km metropolitan grid, where inter-node distances are short enough for both protocols to establish keys at useful rates without range-extending mechanisms. At wider-area scales — city-to-city or national-backbone links — direct and MDI-QKD links become loss-limited and two distinct range-extending architectures become relevant. Trusted-node relay networks forward classical XOR-combined keys across intermediate nodes; this restores range at the cost of end-to-end security, since a compromised relay node exposes the key (Section 5.1). Quantum repeaters, by contrast, extend range through entanglement swapping and distillation, maintaining end-to-end quantum security without trusted intermediaries —

the original proposal [49] showed that nested entanglement purification can overcome the exponential loss/error scaling of a raw quantum channel using only local operations at intermediate stations, without any of the trust assumptions trusted-node relaying requires — but remain a longer-term technology, conventionally requiring quantum memories and mid-circuit measurements at commercially unproven fidelity and scale. All-photonic repeater architectures [50] relax the memory requirement specifically, encoding logical qubits across loss-tolerant photonic clusters instead of storing them, at the cost of substantially higher per-node photon-source and detector counts; this work does not model repeaters of either kind. The case for pursuing them is direct in this work’s own results: the cumulative-configuration decomposition of Section 4 (Figure 9) identifies fibre attenuation as the single dominant term at long range for both protocols — a 3.9% (BB84) and 3.4% (MDI) reduction at 1 km, widening to over 98% for both by 100 km — precisely the loss regime in which a repeaterless link of either protocol becomes impractical, and the regime a network of the scale envisioned in Kimble’s original quantum-internet proposal [51] would need repeaters, of one architecture or the other, to operate in at all. Trusted-node relaying *is* implemented (TBB84, Section 3.1.4) but is exercised only in the real-topology experiments, where range extension is structurally necessary, and deliberately excluded from the core synthetic-grid comparison for the reasons given in Section 5.1. The metropolitan scope of the main comparison isolates the BB84-vs-MDI trade-off from the additional variables introduced by relay-trust forwarding or repeater chain design.

5.4 Implications for Protocol Selection

The results support a protocol-selection framework based on two axes: security requirement and scale. Below the crossover user count N^* (Section 4.3), BB84’s direct mesh is both cheaper and faster than MDI: its bipartite key rate exceeds MDI’s by a roughly constant $\sim 5\times$ under channel-limited conditions (Section 5.1), and its fibre cost undercuts MDI’s despite the steeper $\mathcal{O}(N^2)$ asymptotic scaling. But N^* itself sits only just above the relay count K (Section 4.3), regardless of detector price, so this regime is narrow: for every (N, K) combination actually simulated in this work, N already exceeds N^* , and MDI’s marginal cost of serving each additional user is unconditionally lower than BB84’s (Equation 21), so the cost case for BB84 strengthens only for networks with markedly more relays than users, an unusual regime for a metropolitan deployment — and the relay-count sweep of Section 4.2 shows K itself has a rate-optimal value distinct from its fibre-cost-optimal value (Table 4), so a provider optimising for key rate is pushed towards exactly the larger- K regime that keeps N^* low. Where the deployment scale exceeds N^* , the choice between BB84 and MDI reduces to a security question: MDI eliminates detector side-channel vulnerabilities at the cost of a

lower key rate — a gap that stays close to $\sim 5\times$ under channel-only loss but widens sharply, to over $25\times$, under per-arm hardware degradation such as detector efficiency (Figure 6) — while BB84 delivers higher throughput within the trust assumptions of its detector model.

6 Conclusion

6.1 Summary of Contributions

This work has built a complete simulation and cost modelling infrastructure for the comparative analysis of BB84 and MDI-QKD at metropolitan scale. On the simulation side, both protocols are implemented within the NetSquid discrete-event simulation engine under a ten-configuration cumulative hardware model that progressively introduces physical imperfections from an idealised baseline to a fully realistic system. Point-to-point key rates are obtained via Monte Carlo simulation across a parameter space covering fibre loss, detector efficiency, dark count rate, TX insertion loss, receiver node loss, source error rate, fibre dephasing, detector basis bias, and BSM beam-splitter efficiency. The simulation infrastructure parallelises across CPU cores and persists all results to a structured SQLite database.

At the network level, both protocols are implemented as multi-user simulators: a direct BB84 mesh and an MDI-QKD network with relay BSM stations. Relay positions are placed by a generalised Weiszfeld iteration that minimises total fibre length, and all network experiments are repeated across multiple independent seeds to characterise topology variability. A symbolic cost model expresses total CapEx in terms of free parameters c_s (photon source), c_d (single-photon detector), and c_f (fibre per km), with component counts determined by topology. The model focuses on the structural scaling difference between BB84 and MDI and derives the crossover user count N^* analytically.

Beyond the BB84-versus-MDI comparison itself, this infrastructure was built to generalise. The component/node/protocol separation adopted throughout (Section 3), the cumulative hardware model, the multi-user network simulators, and the SQLite-backed parallel execution pipeline (Appendix E) are not specific to BB84 or MDI-QKD: further protocols, hardware models, and topologies can be added within the same structure. As such, one of this work’s principal contributions is not only the comparative result but early-stage groundwork toward a general-purpose NetSquid-powered QKD network simulator – occupying a role analogous to that of ns-3 [25] or OMNeT++ [26] in classical network simulation. Appendix D identifies specific engineering bottlenecks, chiefly the unrealistic computational cost, that such a tool would need to resolve to reach commercial scale.

6.2 Findings

In the bipartite setting, MDI-QKD’s key rate is structurally lower than BB84’s, a direct consequence of its two-photon BSM coincidence requirement (Section 4.1): roughly a constant

$5\times$ lower under channel-limited conditions such as distance and fibre attenuation, since both protocols share the same link loss, but widening past $25\times$ wherever a per-arm hardware parameter degrades, since such parameters act quadratically on MDI’s coincidence probability but only linearly on BB84’s single-photon detection (Figure 10). Detector efficiency is the dominant *tunable* hardware parameter limiting key rate at short and medium range, while fibre attenuation dominates at long range for both protocols regardless of hardware choice, swallowing over 98% of key rate by 100 km (Figure 9).

This rate gap widens further at the network level through cross-cluster relay-routing detours (Section 5.1). Adding relays helps, but recovers only around a tenth of the BB84 baseline at best, and the recovery is non-monotonic: beyond some point dependent on user count, extra relays fragment the network into smaller catchments faster than they shorten user-to-relay distances (Table 4). The relay count that maximises key rate is also not the one that minimises deployment fibre, so a provider optimising purely for cost and one optimising purely for key rate reach different relay-count decisions even before component prices enter the picture.

The symbolic cost model predicts that MDI becomes cost-competitive above a crossover user count, driven by its linear rather than quadratic fibre scaling with user count (Section 4.3); this crossover is reported as a function of relative detector and fibre cost and network geometry rather than as a single number. Under the assumptions used to derive it, MDI’s marginal cost of serving each additional user is unconditionally lower than BB84’s, and MDI is already the cheaper protocol at every network size simulated in this work.

6.3 Future Work

Several extensions are of immediate interest.

Finite-key corrections. All key rates are asymptotic; finite-key bounds [52] would reduce reported rates and alter the comparison.

Weak coherent pulse sources. Replacing ideal QD sources with WCP laser sources modelled via the decoy-state technique [29] would give rates more directly comparable to published experimental deployments [11, 16, 17] and would substantially reduce assumed source cost.

Asymmetric detector pricing. The cost model (Section 4.3) assumes a single detector price c_d shared by both protocols; a natural extension replaces this with separate c_d^{BB84} and c_d^{MDI} terms and re-derives N^* and the marginal-cost comparison (Equations 18 and 21) as functions of their ratio. This would give a genuine value assessment that the current symmetric model cannot provide.

Cost per unit key rate. Cost and key rate are compared on separate axes throughout this work: Section 4.3 compares total deployment cost at a given (N, K) , while Sections 4.1

and 4.2 compare key rate at a given hardware configuration, and the two are deliberately not combined — the SPAD-BB84-vs-SNSPD-MDI comparison of Section 4.2 explicitly flags that any cost comparison between the two must be read at unequal key rates rather than folded into a single figure. A price-per-bit metric, $C(N, K)/\bar{R}(N, K)$ in \$/kbps, evaluated across the same $(N, K, \text{detector})$ sweeps used throughout this work, would instead give one directly comparable deployment-efficiency number, incorporating the $\sim 5\text{--}25\times$ bipartite key-rate gap (Figure 6) into the cost comparison itself rather than treating it as a caveat alongside it.

Optimal relay placement. k -means relay placement is heuristic. ILP-based placement [22] solves the joint relay placement and routing problem to global optimality, providing tighter lower bounds on minimum MDI deployment cost.

Brownfield relay expansion. The growth-model experiments of Section 3.3 and Appendix B hold the relay count K fixed while users grow, modelling a provider serving unplanned demand on existing infrastructure. The relay count sweep of Section 3.3, by contrast, still re-clusters and repositions all K relays from scratch at each K , which models greenfield network design rather than expanding a live network: a real operator adding capacity has already sunk the cost of its existing K relays and cannot costlessly relocate them. Extending $K \rightarrow K + 1$ realistically would instead fix the existing relay positions and solve for where a single new relay should go — and which existing users it should take over — to minimise the resulting total fibre length. This is a harder problem than the placement methods used elsewhere in this work: the new relay’s optimal position depends on which users it serves, and that assignment depends on its position, so it needs its own alternating position/assignment loop (as in `optimise_relays`) restricted to a single new facility with the other K held fixed, rather than the one-shot strategies used in Appendix B. Not attempted here for want of time.

Relay count sweep at further user counts. The relay count sweep of Section 4.2 showed MDI key rate peaking at $K = 4$ for $N = 15$ (Figure 12b, Table 4) and at $K = 3$ for $N = 10$ (Figure 12a) — consistent with the optimal relay count scaling roughly as $\sqrt{N}$ ($\sqrt{15} \approx 3.9$, rounding to the observed $K = 4$; $\sqrt{10} \approx 3.2$, rounding to the observed $K = 3$), though this rests on only two data points, and the $N = 10$ run has so far been analysed only at the aggregate level, not to the same per-seed rigour as Table 4. A $\sqrt{N}$ relationship would have a plausible geometric basis: for N users spread over a fixed-area service region, dividing into K roughly equal-area catchments shrinks the mean per-catchment spoke radius as $1/\sqrt{K}$ at fixed user density, while the backbone term grows as $K(K - 1)/2$ (Equation 7); balancing a $1/\sqrt{K}$ -shrinking term worth $\mathcal{O}(N)$ against a K^2 -growing term is the kind of trade-off that yields power-law optimal points, though the exact exponent has not been derived analytically here and is offered as a direction rather than a claim. A third data point is needed to say more, and $N = 20$ is a weak choice for it: $\sqrt{20} \approx 4.5$ rounds to 4 or 5 depending on which

side of the boundary the true relationship falls, which does not cleanly separate $\sqrt{N}$ scaling from a simple constant- $K \approx 4$ alternative. A smaller N (e.g. $N = 5$, predicting $K \approx 2$) or a larger one (e.g. $N = 30$, predicting $K \approx 5$) would discriminate the two hypotheses far more sharply, but neither was attempted here; $N = 20$ remains queued regardless, at an estimated ~ 20 hours for $K = 1\text{--}5$ (Table 6), too long to guarantee completion within the project timeline.

Real network topologies. Extending the simulator to accept fixed network graphs from real city infrastructure would enable direct comparison with published deployed networks [11, 16, 17] and provide a practical planning tool for network operators.

Twin-Field QKD. TF-QKD and its variants [32] achieve key rates scaling as $\mathcal{O}(\sqrt{\eta})$, surpassing the PLOB bound and extending the distance at which MDI-level security is achievable without quantum repeaters. Extending this framework to TF-QKD would complete the repeater-less QKD comparison.

Operational expenditure. Incorporating cryogenic OpEx for SNSPD-equipped deployments, using the energetic analysis methodology of [23], would enable full lifetime cost comparisons and may substantially alter the economic case for MDI at high relay counts.

Trusted-node relay networks. At wide-area scales where metropolitan link distances are exceeded and direct MDI-QKD links become loss-limited, trusted-node relay architectures become relevant. Extending the cost model and simulation to trusted-node BB84 would enable comparison across the full distance regime.

Traffic modelling. A model of key request rates, buffer occupancy, and WDM channel sharing would give a more realistic picture of BB84 and MDI at large N , where the direct-mesh assumption becomes economically unphysical.

General-purpose QKD network simulator. The component/node/protocol architecture, cumulative hardware model, and cost-modelling pipeline built here are protocol-agnostic by construction (Section 6). Generalising them into a standalone, documented simulation framework — adding protocol plug-in points beyond BB84 and MDI-QKD, the batched/vectorised trial execution and distributed DES backend discussed in Appendix D, and support for arbitrary network topologies and hardware catalogues — would extend this work from a single comparative study into reusable infrastructure for QKD deployment planning, analogous to the role ns-3 [25] and OMNeT++ [26] play for classical networks.

A Additional Point-to-Point Parameter Sweeps

Section 4.1 discussed four of the eleven point-to-point parameter sweeps produced for this work: distance, detector efficiency, receiver loss, and Charlie’s relay position. Of the remaining seven, three — dark count rate, fibre dephasing rate, and source error rate — are omitted from Figure 14 below and quantified directly in the *Noise-like parameters* discussion instead, since each varies key rate by well under 3% across its realistic hardware range. Figure 14 reproduces the remaining four; the discussion below repeatedly compares these against detector efficiency (Figure 6) and receiver loss (Figure 7) from Section 4.1. In every panel, only the swept parameter deviates from its layer-9 default in Table 2; all other parameters are held at that default operating point (20 km Alice–Bob separation, 0.18 dB/km fibre attenuation, $\eta_d = 0.9$ detector efficiency, 50 cps dark count rate, $L_i = 0.1$ TX insertion loss, 2.0 dB node loss, $\varepsilon_s = 0.015$ source error rate, $\beta = 3.2 \times 10^{-7}$ /km fibre dephasing, $\eta_{bs} = 0.97$ beam-splitter efficiency, Charlie at the arm midpoint), with 1000 Monte Carlo runs per point and shaded bands showing the sample spread.

Three groupings emerge from these seven sweeps, extending the channel-versus-per-arm pattern established in Section 4.1.

Channel-side loss. Fibre attenuation (Figure 14a) behaves exactly like the distance sweep of Section 4.1: the BB84/MDI ratio stays close to $5.2\text{--}5.3\times$ across the full 0–0.3 dB/km range, confirming that fibre attenuation — like distance — acts on the shared channel before Alice’s and Bob’s photons diverge towards Charlie, so it scales both protocols together rather than widening the gap between them.

Per-arm hardware loss. Beam-splitter efficiency (Figure 14b) and TX insertion loss (Figure 14c) both act after the split, and both widen the ratio, consistent with detector efficiency and receiver loss in Section 4.1. Beam-splitter efficiency is unique among all eleven sweeps in that BB84 is structurally unaffected by it (the dashed reference line in Figure 14b is exactly flat, since η_{bs} only enters MDI’s Bell-state measurement); MDI key rate alone declines from 66.1 kbps at $\eta_{bs} = 1.0$ to 42.3 kbps at $\eta_{bs} = 0.8$, a 36% reduction from a single power of η_{bs} in the coincidence probability. TX insertion loss widens the BB84/MDI ratio from $4.8\times$ at $L_i = 0$ to $9.7\times$ at $L_i = 0.5$, the same per-arm doubling pattern seen under detector efficiency (Figure 6) and receiver loss (Section 4.1). X-basis detector efficiency, or basis bias (Figure 14d), belongs to this group too but widens the ratio only modestly, from $5.0\times$ at $\eta_X = 1.0$ to $6.1\times$ at $\eta_X = 0.5$, far less than the full detector-efficiency sweep of Figure 6, because here only one of the two measurement bases is degraded rather than both.

Noise-like parameters. Dark count rate, fibre dephasing rate, and source error rate leave both protocols essentially flat across their full swept ranges at this metropolitan op-

erating point, with the BB84/MDI ratio staying within a few percent of $5.2\times$ throughout each — flat enough that the three are omitted from Figure 14 and reported numerically here instead. Across each parameter’s realistic hardware range, key rate varies from best to worst point by: dark count rate, 0.8% (BB84) and 1.9% (MDI); fibre dephasing rate, 0.6% (BB84) and 2.0% (MDI); source error rate, 0.7% (BB84) and 3.0% (MDI). None of the three introduces a directional loss term on either arm, so none is expected to shift the ratio; as noted in Section 5.3, this flatness is specific to the signal rates achieved at 20 km and should not be extrapolated to longer, detection-starved links.

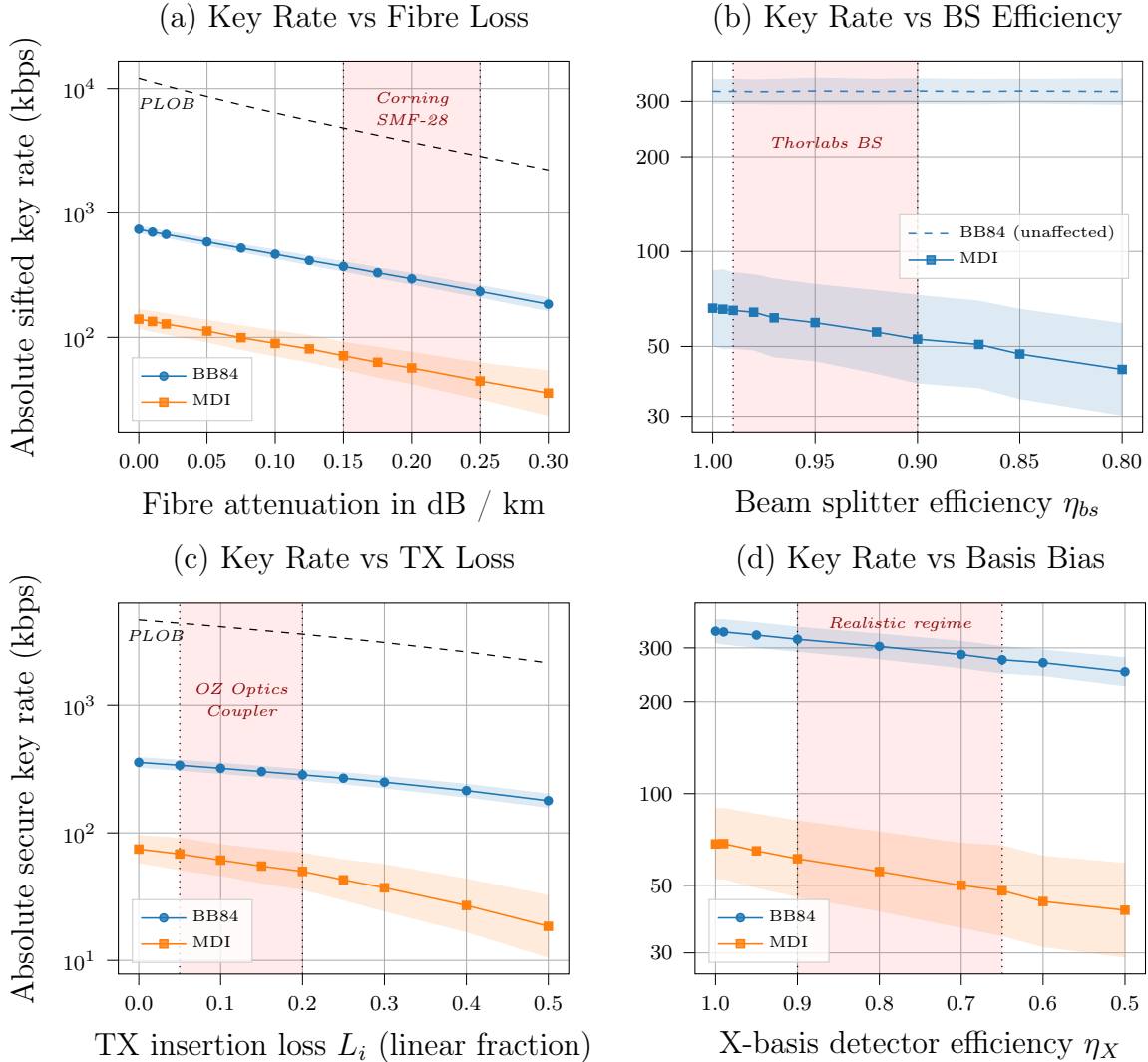


Figure 14: Four of the remaining point-to-point parameter sweeps (see *Noise-like parameters* below for the other three). BB84 in blue (circles), MDI in orange (squares); shaded bands show Monte Carlo sample spread. Except for the parameter swept in each panel, all hardware parameters take their layer-9 default values from Table 2.

B Relay Placement Optimisation

Section 3.1.4 proposed replacing the plain k -means-centroid relay placement used elsewhere in this work’s development with a generalised Weiszfeld iteration, on the grounds that it correctly minimises total fibre length — the network’s actual linear-cost objective — rather than the sum of squared distances that k -means centroids minimise. This appendix works through that argument in more detail: it derives the coupled update of Equation 8 from first principles, works a small example to build intuition for how and why the two objectives diverge, and reports a self-contained geometric comparison of the two placement methods across synthetic topologies matching those used in Section 3.1.4. This comparison is purely geometric — total fibre length under each placement, independent of NetSquid — since, as noted in Section 3.1.4, no network-level key rate results have yet been generated under the new placement method; the discussion questions at the end of this appendix are correspondingly open rather than resolved.

Derivation of the coupled update. This is the classical multi-facility Weber problem, dating to Miehe’s original network-of-junctions formulation [53] and specialised to hub-and-spoke interconnection by O’Kelly [54]; Alumur and Kara [55] survey the broader hub-location literature that has since grown around it. Recall the total fibre length objective from Equation 7,

$$L(x_1, \dots, x_K) = \sum_{k=1}^K \sum_{i \in C_k} \|x_k - p_i\| + \sum_{k < l} \|x_k - x_l\|,$$

where C_k is the fixed set of users assigned to relay k and x_k its free position. Away from the (measure-zero) configurations where x_k coincides with a user p_i or another relay x_l , L is differentiable, and each term contributes a unit vector pointing from the fixed point towards x_k , scaled by the reciprocal distance:

$$\nabla_{x_k} L = \sum_{i \in C_k} \frac{x_k - p_i}{\|x_k - p_i\|^2} + \sum_{l \neq k} \frac{x_k - x_l}{\|x_k - x_l\|^2}.$$

Setting $\nabla_{x_k} L = 0$ and separating the x_k -weighted terms from the fixed-point-weighted terms on each side gives

$$x_k \left(\sum_{i \in C_k} \frac{1}{\|x_k - p_i\|} + \sum_{l \neq k} \frac{1}{\|x_k - x_l\|} \right) = \sum_{i \in C_k} \frac{p_i}{\|x_k - p_i\|} + \sum_{l \neq k} \frac{x_l}{\|x_k - x_l\|},$$

which rearranges directly to the fixed-point update of Equation 8 by evaluating the right-hand side at iteration t and solving for x_k at iteration $t+1$. This is a majorize-minimise step,

not a gradient step: each relay is repositioned to the weighted mean of the points pulling on it, with weights set by the *previous* iterate’s distances, and iterating is guaranteed to be non-increasing in L (Kuhn [41]) even though no step size is chosen explicitly.

The degenerate case, where an iterate $x_k^{(t)}$ lands exactly on a p_i or x_l , produces a division by zero, since the corresponding term is non-differentiable there (the norm has a subgradient of unit magnitude in any direction rather than a unique gradient). In practice, and in the numerical experiments below, this is handled by flooring every pairwise distance at a small ϵ (we use $\epsilon = 10^{-9}$ km) before taking its reciprocal; this perturbs the objective negligibly at realistic network scales but avoids the singularity entirely. A more principled fix — checking the subgradient optimality condition explicitly at any iterate landing within ϵ of a fixed point, as in the standard single-facility Weiszfeld literature — was not needed at the N, K scales explored here, since exact coincidence between a continuous-uniform random draw and another point has probability zero and was never observed in $30 \times 9 = 270$ seeded topology trials.

Worked example: centroid vs. geometric median. A small hand-constructed instance builds intuition for the divergence between the two objectives: eight users in two well-separated groups of four, one of which contains a single point noticeably further from the other three than they are from each other. The k -means centroid of that cluster is pulled towards this one distant point, since it enters the sum-of-squares objective quadratically; the geometric median instead settles closer to the tight group of three, since the same point enters the sum-of-*unsquared*-distances objective only linearly. This is the standard robustness property of the median relative to the mean, familiar from one-dimensional statistics, applied here to a two-dimensional facility-location setting. The effect on total fibre length is modest but directionally consistent with every trial in the larger comparison below: 15.53 km under the k -means centroid versus 14.62 km under the coupled Weiszfeld update, a 5.9% reduction, obtained without changing which users are assigned to which relay (cluster assignment is identical under both placements).

Geometric comparison across network topologies. To quantify the fibre-length reduction at realistic network scales, both placement methods were run on synthetic topologies generated identically to those used in Section 3.1.4: N users placed uniformly at random within a 10×10 km service area, for $N \in \{10, 20, 50\}$ and relay counts $K \in \{2, 3, 5\}$, over 30 independent seeds per (N, K) pair. For each seed, k -means clustering ($n_{\text{init}} = 10$) provides both the baseline centroid placement and the initial cluster assignment for the coupled Weiszfeld refinement (Equation 8), which then alternates position updates with nearest-relay reassignment until total fibre length changes by less than 10^{-9} km between iterations (typically several dozen iterations; up to 146 in the smallest, most relay-dense case). Table 5 reports

total fibre length — spoke plus backbone, Equation 7 — under each method, averaged across seeds.

N	K	k -means length (km)	Weiszfeld length (km)	Reduction
10	2	29.95	28.57	4.6% $\pm$ 2.7
10	3	35.42	30.92	12.6% $\pm$ 3.8
10	5	69.58	37.89	45.6% $\pm$ 3.7
20	2	57.23	56.00	2.1% $\pm$ 1.4
20	3	56.31	53.61	4.8% $\pm$ 1.8
20	5	84.80	67.13	20.8% $\pm$ 4.6
50	2	143.01	142.11	0.6% $\pm$ 0.4
50	3	124.03	122.06	1.6% $\pm$ 0.8
50	5	133.63	126.01	5.7% $\pm$ 1.9

Table 5: Mean total fibre length (spoke + backbone) under k -means centroid versus generalised Weiszfeld relay placement, across 30 seeds per (N, K) pair, 10×10 km service area. Reduction is $100(L_{\text{kmeans}} - L_{\text{Weiszfeld}})/L_{\text{kmeans}}$, mean $\pm$ one standard deviation across seeds.

The reduction is positive in every trial — consistent with the monotonic non-increase guarantee noted above, since the Weiszfeld iteration starts from the k -means solution and never leaves it worse off — and grows sharply with K at fixed N : from 4.6% at $K = 2$ to 45.6% at $K = 5$ for $N = 10$, and from 0.6% to 5.7% across the same K range for $N = 50$. This follows directly from Equation 7: the backbone term has $K(K - 1)/2$ pairwise contributions, growing quadratically in K , while the spoke term grows only linearly in N ; at larger K the backbone increasingly dominates total cost, and it is precisely the backbone coupling that a per-cluster centroid ignores entirely. Conversely, the reduction shrinks as N grows at fixed K — from 45.6% to 5.7% at $K = 5$ between $N = 10$ and $N = 50$ — because a larger user base dilutes each relay’s positioning error relative to the now spoke-dominated total, and because tighter, more populated clusters leave less room for a single geometric outlier to pull the centroid away from the coupled optimum. The practical implication is that this refinement matters most exactly where the MDI network topology already carries a real relay count K relative to N — the regime the cost-crossover analysis of Section 4.3 identifies as commercially relevant for smaller deployments before BB84’s $\mathcal{O}(N^2)$ fibre term dominates — and least where N is large enough that per-user fibre cost already dominates the network total regardless of relay placement.

Network-level comparison under organic growth. The geometric comparison above holds cluster assignment and N fixed. A second, network-level comparison (`relay_placement.py`) instead asks how different placement strategies compare as a network grows under the fixed-

membership provider-growth model of Section 3.3: $N_{\min}$ users are drawn into K fixed catchments, each relay is placed once by the strategy under test, and the relay position is then *frozen* while further users join one at a time, each assigned to a uniformly-random existing catchment. Three strategies are compared: *centroid*, the mean position of a catchment’s users; *boundary*, the centroid displaced by one standard deviation of the catchment’s own spread, in the direction of the population-weighted mean of the other catchments’ centroids (a heuristic proxy for the backbone term, without solving for it); and *Weiszfeld*, Equation 8 with catchment membership held fixed rather than reassigned. Experiment 1 runs this comparison at $K = 1$ (a single catchment spanning the whole service area), where only centroid and Weiszfeld apply — boundary is undefined without a second catchment to displace toward, and Weiszfeld reduces to the plain single-facility geometric median of Equation 6 since the backbone term in Equation 8 vanishes at $K = 1$. Experiment 2 runs it at $K = 2$, comparing all three strategies. Because the growth sequence, catchment split, and catchment membership are identical across strategies for a given seed — only where each strategy chooses to place its relay at $N_{\min}$ differs — the comparison isolates the effect of the initial placement decision on robustness to unplanned subsequent growth, rather than re-answering the already-resolved question of which placement is instantaneously optimal for a fixed user set.

Open questions. The comparison above resolves the geometric question — which placement minimises total fibre length for a *fixed* backbone topology and cluster assignment — but leaves several modelling questions open, appropriate for discussion before the new placement method is adopted for the network-level key rate sweeps of Section 3.1.4:

- *Backbone topology.* Table 3 assumes a full $K(K - 1)/2$ relay mesh. A minimum spanning tree or Steiner-tree backbone would in general have strictly lower total length for $K > 2$, and would change the placement problem itself (the coupling term in Equation 7 would run over tree edges rather than all pairs). Whether the full mesh is retained for redundancy/latency reasons independent of cost, or replaced with a cheaper backbone topology, should be decided before the cost model and placement method are finalised together.
- *Assignment optimality.* Cluster reassignment after each position update is by nearest Euclidean distance to a relay, which is the correct assignment rule for the spoke term alone but does not account for the backbone term: a user could in principle be assigned to a farther relay if doing so allowed both relays to move to a jointly cheaper configuration. This is a combinatorial partitioning question distinct from the continuous position optimisation solved by Equation 8, and was not explored here.

- *Convergence cost.* The Weiszfeld refinement took up to 146 outer iterations to reach a 10^{-9} km convergence tolerance at the smallest, most relay-dense configuration tested ($N = 10$, $K = 5$); this is negligible against NetSquid’s own per-trial cost (Appendix D) but would be worth profiling at the $N = 100$ scale flagged there as currently infeasible.
- *Net effect on key rate.* A shorter mean path length should translate into a modest key rate improvement through reduced fibre attenuation (Section 3.1.2, layer 1), but by how much depends on where in the loss-versus-distance curve of Figure 9 the affected links sit; this can only be answered by re-running the network-level sweeps of Section 3.1.4 under the new placement, which had not yet been done at the time of writing.
- *Degenerate coincidence.* Both Equations 6 and 8 divide by the distance between the current iterate and each fixed point, which is undefined if a relay iterate ever coincides exactly with a user (or, in the multi-facility case, another relay). The implementation guards against this by flooring the distance at 10^{-9} km rather than applying Ostresh’s modification — the standard fix, which tests whether the coincident point is itself already a stationary point of the weighted sum and, if not, steps away from it along a descent direction. The floor avoids a division-by-zero crash but is not equivalent: it lets that term’s weight grow to $\mathcal{O}(10^9)$, which can pin the next iterate back near the same point rather than resolving the singularity correctly. Coincidence is measure-zero under the continuous uniform and Gaussian user placements used throughout this work, so it was not observed to trigger in practice, but the implementation should not be trusted as-is on discretised or clustered location data (e.g. users snapped to a road network, or a relay initialised exactly at a fixed real-topology co-ordinate) where exact coincidence is plausible.
- *Which topologies favour Weiszfeld.* Weiszfeld does not uniformly beat centroid or boundary placement at the network-key-rate level: across the growth-model comparison above, its ranking against the other two strategies varies from seed to seed and from N to N , sometimes best, sometimes not. This is expected in principle — centroid minimises spoke cost per catchment in isolation, while Weiszfeld additionally couples in the backbone term, so the two objectives should diverge most when the backbone is a non-negligible fraction of total fibre length (catchments close together relative to their spread, or markedly unequal in size) and coincide when it is not. Since catchment membership for $N > N_{\min}$ is now assigned by an independent uniform coin flip per new user rather than a fixed even split, catchment size imbalance is a first-class, randomly-varying outcome of the growth process itself rather than a fixed property of the initial topology — strengthening the case that it is a real driver of the mixed

ranking, though this work only observed the mixed outcome without identifying which geometric property of a given seed predicts it. A natural follow-up would be to regress the per-seed key rate gap between strategies against candidate covariates (cluster size imbalance realised by $N_{\max}$, inter-cluster separation relative to intra-cluster spread, backbone-to-spoke cost ratio at the initial placement) using the per-seed data already logged by `relay_placement.py`, and, if a clear predictor emerges, confirm it with a dedicated sweep that varies that one covariate while holding the others fixed. Not undertaken here for want of time.

C Real-Topology Case Study: The BT London Network

Every network-scale result in the main text uses synthetic user and relay placements, uniformly or Gaussian-distributed over an idealised square service area. As a preliminary check that the qualitative conclusions of Section 4.3 survive contact with a genuine, geographically real topology, `network/real_topology.py` loads fixed relay coordinates from real-world sites and projects them to a local metropolitan grid (equiarectangular projection, accurate to under 1% at this scale). The topology used here, `data/real_topologies/bt.json`, fixes $K = 3$ relays at three real London telecommunications sites — Slough, the West End, and the City of London — while users are drawn from the same Gaussian catchment model used throughout the main text, now centred on these real coordinates rather than a synthetic grid.

This case study additionally uses an earlier, fully numeric GBP-denominated cost model (`network/cost.py`, superseded during this work by the symbolic, price-agnostic treatment of Section 4.3 once it became clear manufacturer prices could not be pinned down precisely enough to justify absolute figures). Unlike the symbolic model, it includes passive relay hardware cost directly — the HOM beam splitter, polarising beam splitters, and optical switch that Section 4.3 explicitly omits for simplicity — at illustrative prices of a custom $\eta_d = 0.90$ detector (£107k, linearly interpolated between a £15k SPAD at $\eta_d = 0.20$ and a £100k SNSPD at $\eta_d = 0.85$), a £150k QD source, and £10k/km of installed fibre. These figures are illustrative anchors, not validated market prices, and the results below come from a single random seed; they are offered as a concrete, real-topology sanity check on the analytical results above, not as a replacement for them.

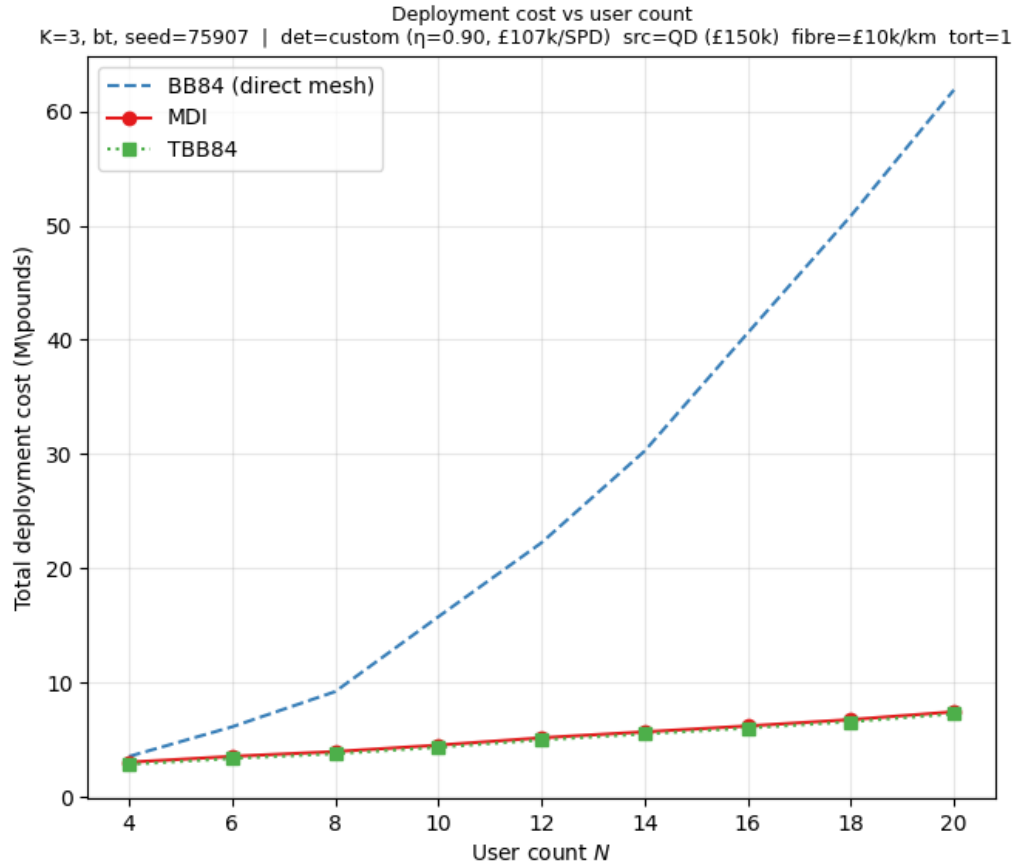


Figure 15: Total deployment cost against user count N for the BT London topology ($K = 3$). BB84’s direct mesh diverges quadratically; MDI and TBB84 stay low and nearly coincide, consistent with the N^* ceiling of at most $K + 1 = 4$ users derived analytically in Section 4.3 for this K .

Figure 15 reproduces, on a real topology and with real-ish component prices, the crossover behaviour derived symbolically in Section 4.3: BB84 is cheapest only at the smallest user counts tested, and by $N = 20$ its cost has grown to roughly eight times MDI’s or TBB84’s, both of which track each other closely throughout.

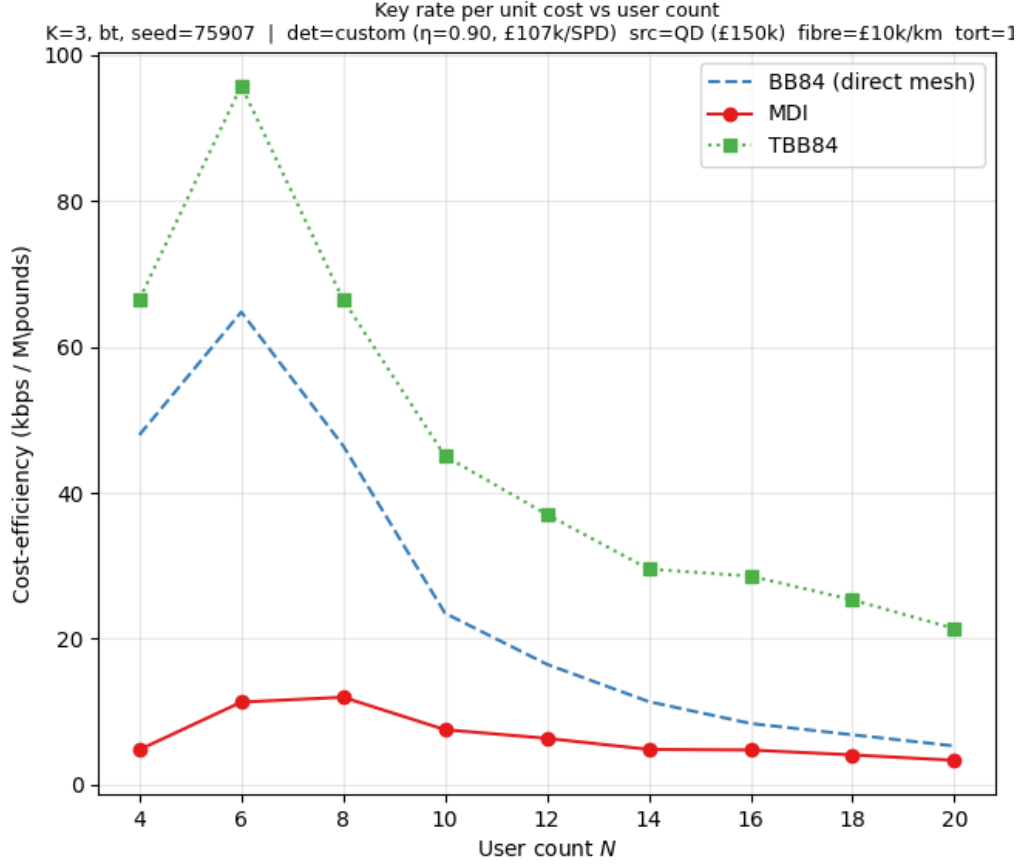


Figure 16: Key rate per unit deployment cost (kbps/£M) for the BT London topology. TBB84 and BB84 both substantially outperform MDI on this metric, since MDI’s lower absolute cost (Figure 15) is more than offset by its much lower key rate (Section 5.1).

Figure 16 is a preliminary instance of exactly the cost-per-unit-key-rate metric proposed as future work in Section 6.3: it inverts the picture painted by total cost alone. MDI is the cheapest protocol at every N tested (Figure 15), but is the *least* cost-efficient once its key-rate deficit is accounted for, since the $\sim 5\text{--}25\times$ bipartite rate gap of Section 5.1 enters the denominator directly. This is a single-seed, single-topology result and should not be read as a general ranking, but it demonstrates that total cost and cost-efficiency can disagree sharply, which total-cost figures alone — including Section 4.3’s own headline result — cannot show on their own.

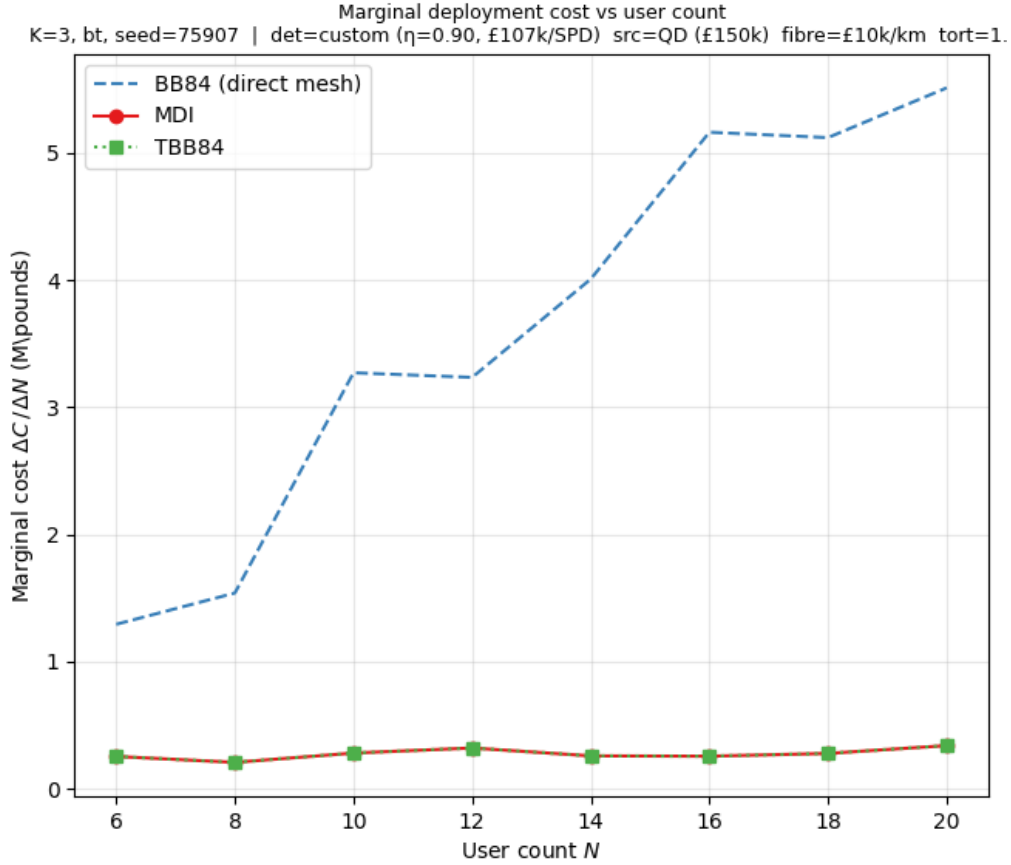


Figure 17: Marginal deployment cost, $\Delta C/\Delta N$, for the BT London topology. BB84’s marginal cost rises with N ; MDI’s and TBB84’s stay flat and low throughout.

Figure 17 is an independent empirical measurement of exactly the qualitative result proved in closed form in Equation 21: BB84’s marginal cost of adding a user grows with network size (each new user needs a direct link to every existing one) while MDI’s stays flat (a new user needs only one spoke to an already-provisioned relay). By $N = 20$, BB84’s marginal cost is more than an order of magnitude above MDI’s or TBB84’s here, on a real topology with a fuller (non-symbolic) cost accounting — a substantially larger empirical gap than the illustrative single-parameter numbers used to demonstrate Equation 21 in the main text.

TBB84 tracks MDI closely on all three figures despite its different trust model (Section 4.3), since both share the same qualitative cost structure — a small number of relay-side components serving many spoke users — differing mainly in what they require the customer to trust (Section 4.3), not in their fibre or hardware cost scaling. Extending this case study to further real UK topologies, multiple seeds, and the symbolic cost model of Section 4.3 is a natural direction for the future work already flagged there.

D NetSquid: Limitations and Speedup Strategies

Section 5.3 flags NetSquid’s computational cost as a limitation that constrained the seed and Monte Carlo budgets feasible within the project timeline. This appendix details that cost, the mitigation strategies attempted during this work, and the further engineering a commercial-scale successor would plausibly need.

The i.i.d. modelling assumption. The simulations in this work model each photon trial as an independent, identically distributed (i.i.d.) draw: no channel memory, no inter-trial correlations, no time-varying network state. Under this assumption, the key rate distributions produced by NetSquid are in principle reproducible by a simple analytical probability model operating on the same physical parameters, and NetSquid’s overhead — launching a DES engine, propagating quantum state objects, managing event queues — is not strictly necessary for the i.i.d. regime. The justification for using NetSquid regardless is extensibility: the DES framework supports temporally correlated dynamics (detector deadtime, jitter, burst traffic, entanglement distillation protocols) that an i.i.d. model cannot accommodate, and these effects become important if this infrastructure is extended to quantum repeater chains or adaptive network protocols (Section 6.3).

Wall-clock cost. The computational cost of the DES approach is non-negligible: network sweeps across multiple seeds and user counts required on the order of hours of wall-clock time on a multi-core workstation. This constrained the number of seeds and Monte Carlo runs feasible within the project timeline, and limits the statistical confidence of the topology-averaged results. Table 6 summarises representative wall-clock costs on a multi-core workstation: a full point-to-point parameter sweep completes in minutes, but cost grows sharply with network size, reaching hours for a $N=20$ relay sweep and an estimated weeks for $N=100$ — infeasible within the project timeline. The dominant cost drivers are the $\mathcal{O}(N^2)$ user-pair count, the sequential causal event loop within each NetSquid run (which cannot be parallelised across a single trial), and multiprocessing spawn overhead of approximately 2–3 s per Monte Carlo batch, independent of workload.

Scenario	Wall-clock time	Feasibility
P2P sweep	~minutes	OK
$N=20$ relay sweep	~hours	Marginal
$N=100$ sweep	~weeks	Infeasible

Table 6: Wall-clock time for representative NetSquid sweep scenarios on a multi-core workstation.

Mitigation strategies attempted. Two mitigations were investigated during this work. Adaptive Monte Carlo sampling terminates trials early once the running key-rate estimate has converged, reserving the full sample budget for slower-converging points; this reduced BB84 wall-clock time by 30% (28% fewer runs, $<1\%$ relative error) but gave negligible savings for MDI, whose structurally higher variance (Section 4.1, *Sample spread*) rarely triggers early stopping. Gaussian process surrogate modelling instead trains a regression model on a coarse NetSquid-generated grid and evaluates a dense 10^4 -point grid in under 100 ms, giving near-instant interpolation once trained; however, it requires a substantial upfront training sweep, its accuracy degrades outside the training range, and it defers rather than reduces the underlying simulation cost. Neither approach addresses the root cause, since cost is set by control flow and process management (the sequential event loop and spawn overhead) rather than floating-point throughput — for the same reason, GPU acceleration was considered and rejected, as BB84/MDI operate on 1–2-qubit density matrices (2×2 or 4×4) too small for GPU parallelism to outweigh its overhead, and NetSquid has no GPU backend regardless. These constraints motivated the relay-count and user-count ranges explored in Section 3.1.4.

Proposed speedup strategies. The bottlenecks identified above — multiprocessing spawn overhead and a sequential per-trial event loop that cannot be parallelised within a single run — are engineering problems rather than fundamental limits of the discrete-event approach, and none of the following was attempted here for want of time. Batched or vectorised trial execution would amortise spawn cost and interpreter startup across many trials rather than paying it once per Monte Carlo batch. A persistent worker pool that keeps interpreter and NetSquid state warm across sweep points would remove the 2–3 s per-batch spawn overhead entirely, in place of the spawn-per-batch model used throughout this work. For the sequential per-trial event loop itself, a shared-memory or distributed DES backend could parallelise causally independent sub-chains of a single network-scale trial (e.g. non-interacting user pairs), rather than only parallelising across whole trials as done here. Classical network simulation faced an analogous scaling problem before tools such as ns-3 [25] and OMNeT++ [26] matured into viable commercial-scale planning tools; the wall-clock costs quantified in Table 6 are offered as a first characterisation of where NetSquid’s current

execution model breaks down, for a comparable QKD-specific tool to build on. Read this way, the open question raised by this appendix is not whether NetSquid’s general-purpose discrete-event engine is the right abstraction for network-scale QKD simulation in principle, but what engineering investment is required to make it practical at the network sizes a commercial deployment-planning tool would need to support.

E Code and Reproducibility

Availability. All simulation, network, cost-model, and plotting code is publicly available at <https://github.com/linuskelsey/netsquid-qkd>, released under a permissive MIT-style licence. The repository is organised into `BB84/` and `MDI/` (protocol implementations and simulation runners), `network/` (the multi-user topology layer and BB84/MDI/trusted-BB84 network simulators), `lib/` (shared configuration loading, database, and plotting utilities), `configs/` (the ten JSON hardware-layer presets of Table 2), `scripts/P2P/` and `scripts/network/` (the point-to-point and network-scale sweep drivers behind every figure in this report), `scripts/analyse/` (figure reconstruction from stored results, described below), and `docs/` (this dissertation’s $\text{\LaTeX}$ source and figures). Every script exposes a `-help` option and carries a usage docstring; running any figure-producing script end to end requires only `pip install -r requirements.txt` plus a free academic registration with NetSquid itself at netsquid.org, since NetSquid is not distributed on the public PyPI index.

Results database. Every Monte Carlo batch — whether from a point-to-point sweep, a network-scale sweep, or an ad-hoc script invocation — is persisted to a local SQLite database (`data/results.db`) via `lib/db.py`, rather than being discarded once a figure is drawn. Two tables cover all simulation output: `p2p_results`, one row per Monte Carlo run of a single Alice–Bob (or user–relay) exchange, recording every swept hardware parameter alongside the resulting key rate, QBER, and status; and `network_results`, one row per network-scale simulation call (one protocol, one seed, one sweep point), joinable back to its constituent `p2p_results` rows via `net_run_id`. The full schema is documented in `DATA.md`. This design decouples analysis from re-simulation: `scripts/analyse/` can reconstruct any figure in this report directly from the database, re-running only the aggregation and plotting code, which is what made it practical to revisit plot formatting, error-display mode, and shaded-region annotations (Section 4.1) without re-incurring the wall-clock costs of Appendix D. The database itself is not version-controlled (`data/` is git-ignored): it is regenerated by construction on first run and grows unboundedly as further sweeps are executed, so committing it would mean committing an ever-growing, non-reviewable binary artefact rather than the code that produces it. The raw database backing the results in this report is available on request.

Figure provenance. Every figure in this report also has its full provenance preserved under `docs/figures/August/final-figs/` in the repository. Each figure’s directory contains an `assumptions.md` file recording every fixed parameter used to produce it (seed, user and relay counts, service area, tortuosity, runtimes per pair, and the resulting key rates), alongside the rendered plot and its $\text{\LaTeX}$ source. Network-scale figures additionally include a `topologies/`

subdirectory with the exact relay and user placement underlying each data point — one image per relay count and protocol — so the specific geometry behind any reported key rate can be inspected directly rather than only its aggregate statistics.

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